SDS 4030 - Statistics for Data Science II: Final Project

Main Markdown File

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Preliminary library loading, reading code in, and exploratory data analysis

```r
gss = read.table("gss.txt", header = TRUE)
summary(gss)
```

```
##       Age             Educ           Hours           Sex
##  Min.   :18.00   Min.   : 4.00   Min.   : 3.00   Length:815
##  1st Qu.:32.00   1st Qu.:12.00   1st Qu.:39.00   Class :character
##  Median :42.00   Median :14.00   Median :40.00   Mode  :character
##  Mean   :42.91   Mean   :13.93   Mean   :42.59
##  3rd Qu.:52.00   3rd Qu.:16.00   3rd Qu.:50.00
##  Max.   :88.00   Max.   :20.00   Max.   :89.00
##     TVHours          Happy               MEduc           FEduc
##  Min.   : 0.000   Length:815         Min.   : 0.00   Min.   : 0.00
##  1st Qu.: 2.000   Class :character   1st Qu.:10.00   1st Qu.: 9.00
##  Median : 2.000   Mode  :character   Median :12.00   Median :12.00
##  Mean   : 2.709                      Mean   :11.61   Mean   :11.45
##  3rd Qu.: 3.000                      3rd Qu.:14.00   3rd Qu.:14.00
##  Max.   :20.000                      Max.   :20.00   Max.   :20.00
##    Marital               Sibs            Work              Zodiac
##  Length:815         Min.   : 0.000   Length:815         Length:815
##  Class :character   1st Qu.: 2.000   Class :character   Class :character
##  Mode  :character   Median : 3.000   Mode  :character   Mode  :character
##                     Mean   : 3.439
##                     3rd Qu.: 5.000
##                     Max.   :16.000
##      Income
##  Min.   :     500
##  1st Qu.:   23750
##  Median :   45000
##  Mean   :  324767
##  3rd Qu.:   67500
##  Max.   :28637626
```

We begin our data science workflow by familiarizing ourselves with the GSS dataset. The dataset contains 815 rows corresponding to 815 individuals' survery responses. The dataset contains 13 variables. There are 8 discrete numerical variables: Age (Responder's Age), Educ (Total years of education), Hours (Weekly work hours), TVHours (Hours spent watching TV daily), MEduc (Mother's

years of education), FEduc (Father's years of education), Sibs (Number of siblings), and Income (Yearly income). Thus there are 5 categorical variables: Sex (Responder's sex), Happy (Happiness level), Marital (Marital status), Work (Work status), and Zodiac (Zodiac sign). Uniquely they take on the following values: Sex (Female, Male), Happy (Pretty Happy, Very Happy, Not Too Happy), Marital (Divorced, Widowed, Married, Never Married, Separated), Work (Fulltime, Housekeeping, Laid Off, School, Retired, Parttime, Not Working, Other), and Zodiac (Virgo, Libra, Taurus, Pisces, Cancer, Aquarius, Leo, Capricorn, Aries, Gemini, Sagittarius, Scorpio).

Now that we understand our data we isolate our goal being to build a linear model for the response variable yearly income by treating all other seven quanitative variables as covariates and ignoring all five qualitative variables.

Step 1:

i) Fit a multiple linear model with all available quantitative covariates.

ii) Perform exact and asymptotic tests for the set of linear constraints which has been specified for your team.

iii) Do all of your conclusions agree with each other and what are the relationships between the different tests?

---

i) Fitting a multiple linear model with all available quantitative covariates. This is the model that we will use for the next few parts of this project

```r
library(faraway)
gss <- gss[gss$Income < 2e+05, ]

full_model <- lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
    Sibs, gss)
summary(full_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -60108 -20318  -4862  15321  93891
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -16351.02    8247.23  -1.983 0.047757 *
## Age            150.46      74.32   2.024 0.043264 *
## Educ          3364.59     424.80   7.920 8.03e-15 ***
## Hours          180.47      70.77   2.550 0.010962 *
## TVHours      -1687.84     508.22  -3.321 0.000938 ***
## MEduc         -251.63     386.81  -0.651 0.515539
## FEduc          790.44     341.55   2.314 0.020908 *
## Sibs            75.97     399.52   0.190 0.849237
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27610 on 790 degrees of freedom
## Multiple R-squared:  0.1522, Adjusted R-squared:  0.1447
## F-statistic: 20.26 on 7 and 790 DF,  p-value: < 2.2e-16
```

For the sake of later model comparison we will find the least squares estimate of the coefficient vector, an unbiased estimate of the residual variance and the MLE of the residual variance in the full model.

```r
Y = gss$Income
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2]
beta = solve(crossprod(X), crossprod(X, Y))[, ]
print(beta)
```

```
##  (Intercept)          Age         Educ        Hours      TVHours        MEduc
## -16351.01822    150.46215   3364.58602    180.46714  -1687.84258   -251.63437
##        FEduc         Sibs
##    790.44085     75.97103
```

```r
residuals = Y - X %*% beta
sigma = sqrt(mean(residuals^2))
print(sigma^2)
```

```
## [1] 754521699
```

```r
S = sqrt(sum(residuals^2)/(n - p))
print(S^2)
```

```
## [1] 762162425
```

Our full model is $Y_i = \beta_0 + \beta_{Age}\text{Age}_i + \beta_{Educ}\text{Educ}_i + \beta_{Hours}\text{Hours}_i + \beta_{TVHours}\text{TVHours}_i + \beta_{MEduc}\text{MEduc}_i + \beta_{FEduc}\text{FEduc}_i + \beta_{Sibs}\text{Sibs}_i + \epsilon_i$. An unbiased estimate of the residual variance is

equal to 5.42076e+12 and the MLE of the residual variance is equal to 5.474497e+12, under the full model.

The reduced model is a little more tricky where we have to create a model in which it follows the constraints given by: $H_0 : 10 * \beta_{MEduc} = 20 * \beta_{FEduc} = 2$.

We can rewrite the constraint as $10 * \beta_{MEduc} = 2$, $20 * \beta_{FEduc} = 2$. Or, $\beta_{MEduc} = 0.2$, $\beta_{FEduc} = 0.1$

We can plug the null values into the full model, replacing coefficients with the constants so the full model becomes $Y_i = \beta_0 + \beta_{Age}\text{Age}_i + \beta_{Educ}\text{Educ}_i + \beta_{Hours}\text{Hours}_i + \beta_{TVHours}\text{TVHours}_i + 0.2 * \text{MEduc}_i + 0.1 * \text{FEduc}_i + \beta_{Sibs}\text{Sibs}_i + \epsilon_i$

We have to set MEduc and FEduc, the fixed parts of the response in the reduced model, as the offset.

```r
reduced_model <- lm(Income ~ Age + Educ + Hours + TVHours + Sibs, gss,
    offset = 0.2 * MEduc + 0.1 * FEduc)
summary(reduced_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + Sibs, data = gss,
##     offset = 0.2 * MEduc + 0.1 * FEduc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -64499 -20150  -3907  15527  87703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -10203.68    7408.22  -1.377 0.168794
## Age             99.38      69.38   1.432 0.152398
## Educ          3603.18     397.30   9.069  < 2e-16 ***
## Hours          173.62      70.88   2.450 0.014518 *
## TVHours      -1752.72     508.36  -3.448 0.000595 ***
## Sibs          -125.04     388.09  -0.322 0.747381
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27670 on 792 degrees of freedom
## Multiple R-squared:  0.1459, Adjusted R-squared:  0.1406
## F-statistic: 27.07 on 5 and 792 DF,  p-value: < 2.2e-16
```

```r
R <- rbind(c(0, 0, 0, 0, 0, 1, 0, 0), c(0, 0, 0, 0, 0, 0, 1, 0))
# the 1's in R represent the MEduc and FEduc since those are within
# the null hypothesis
r <- c(0.2, 0.1)
k <- length(r)
beta0 = beta - solve(crossprod(X), crossprod(R, solve(R %*% solve(crossprod(X),
```

```r
    t(R)), R %*% beta - r)))[, ]
print(beta0)
```

```
##  (Intercept)          Age         Educ        Hours      TVHours        MEduc
## -10203.68500     99.38236   3603.18025    173.62023  -1752.72393      0.20000
##        FEduc         Sibs
##      0.10000   -125.04415
```

```r
residuals0 = Y - X %*% beta0
sigma0 = sqrt(mean(residuals0^2))
print(sigma0^2)
```

```
## [1] 760056321
```

```r
S0 = sqrt(sum(residuals0^2)/(n - p + k))
print(S0^2)
```

```
## [1] 765814323
```

An unbiased estimate of the residual variance is equal to 5.421657e+12 and the MLE of the residual variance is equal to 5.461868e+12, under the reduced model.

ii) Perform exact and asymptotic tests for the set of linear constraints which has been specified for your team.

We have to test the hypothesis $H_0 : 10 * \beta_{MEduc} = 20 * \beta_{FEduc} = 2$ against exact and asymptotic tests. The best exact test is, of course, the exact F-test assuming we are have normalcy conditions (which we will check later, but step 1 ensures that we do it regardless of the normalcy checks). To test for the asymptotic tests it is logical to use the Wald test since we have a large sample (n = 815). It is also one of the easier asymptotic tests to check, so we will use it due to its efficiency, accuracy, and sample size.

Exact F- Test

```r
FT = crossprod(R %*% beta - r, solve(R %*% solve(crossprod(X), t(R)),
    R %*% beta - r))[, ]/(k * S^2)
print(FT)
```

```
## [1] 2.897433
```

```r
pvalue = pf(FT, k, n - p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.05575113
```

```r
anova(reduced_model, full_model)
```

```
## Analysis of Variance Table
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   Res.Df        RSS Df  Sum of Sq      F  Pr(>F)
## 1    792 6.0652e+11
## 2    790 6.0211e+11  2 4416628562 2.8974 0.05575 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The observed value of the F test statistic is equal to 0.06680263 and the corresponding p-value is equal to 0.935385, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

Wald Test

```r
WT = crossprod(R %*% beta - r, solve(R %*% solve(crossprod(X), t(R)),
    R %*% beta - r))[, ]/S^2
print(WT)
```

```
## [1] 5.794865
```

```r
pvalue = pchisq(WT, k, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.05516467
```

```r
anova(reduced_model, full_model, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   Res.Df        RSS Df  Sum of Sq Pr(>Chi)
## 1    792 6.0652e+11
## 2    790 6.0211e+11  2 4416628562  0.05516 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The observed value of the Wald test statistic is equal to 0.1336053 and the corresponding p-value is equal to 0.9353798, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$

Score Test

```
SF = crossprod(X, residuals0)/sigma0^2
FI = crossprod(X)/sigma0^2
ST = crossprod(SF, solve(FI, SF))[, ]
print(ST)
```

```
## [1] 5.810923
```

```
P = X %*% solve(crossprod(X), t(X))
ST = sum((P %*% residuals0)^2)/sigma0^2
print(ST)
```

```
## [1] 5.810923
```

```
pvalue = pchisq(ST, k, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.05472354
```

The observed value of the Score test statistic is equal to 0.1349074 and the corresponding p-value is equal to 0.934771, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

Likelihood Ratio Test

```
LR = -2 * (logLik(reduced_model)[1] - logLik(full_model)[1])
print(LR)
```

```
## [1] 5.832183
```

```
LR = n * log(sigma0^2/sigma^2)
print(LR)
```

```
## [1] 5.832183
```

```r
pchisq(LR, k, lower.tail = FALSE)
```

```
## [1] 0.05414489
```

```r
library(lmtest)
lrtest(reduced_model, full_model)
```

```
## Likelihood ratio test
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   7 -9291.4
## 2   9 -9288.5  2 5.8322    0.05414 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The observed value of the Likelihood Ratio test statistic is equal to 0.1349186 and the corresponding p-value is equal to 0.9347658, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

iii) Do all of your conclusions agree with each other and what are the relationships between the different tests?

Yes, all of my conclusions agree with each other as in the exact F test, Wald test, score test, and likelihood ratio test we have failed to reject the null, and they all yield similar p-values. Both the F-Exact test and asymptotic tests provide very small test statistics which means we do not have substantial evidence to reject the null hypothesis.

```r
WT = k * FT
print(WT)
```

```
## [1] 5.794865
```

```r
ST = WT * S^2/sigma0^2
print(ST)
```

```
## [1] 5.810923
```

```r
LR = n * log(n/(n - p) * WT/ST)
print(LR)
```

```
## [1] 5.832183
```

```r
LR = n * log(1 + WT/(n - p))
print(LR)
```

```
## [1] 5.832183
```

```r
LR = -n * log(1 - ST/n)
print(LR)
```

```
## [1] 5.832183
```

In the F test, the exact F-statistic is 0.06680263, and the p-value is 0.935385. In the Wald test the Wald statistic is 0.1336053 and the p-value is 0.9353798. According to Slutsky's theorem, we infer that W_n = kF, the Wald statistic equals kF asymptotically. The Wald statistic is k times the 0.06680263 where k=2. While the F test relies on a finite sample, the Wald statistic follows a chi-square distribution and asymptotic approximations.

In the score test, the score test statistic is 0.1349074 and the p-value is 0.934771 We know that $\hat{\sigma}_0^2 ST = S^2 W_n$. The Wald and Score test statistics only differ in terms of the residual variance estimation. The Wald test statistic utilizes an unbiased estimator of the residual variance under the full model while the Score test statistic utilizes the MLE of the residual variance under the reduced model. In other words the Wald test uses the full model variance and the Score test uses reduced model variance. Thus we can see how the two test statistics differ slightly.

In the likelihood ratio test, likelihood test statistic is 0.1349186 and the p-value is 0.9347658. Below we see how the Likelihood test statistic relates to the Wald test and Score test through transformations, implying they provide the same inference for very large sample sizes.

So yes, we observe that the results of all the four test statistics concur. The results of the asymptotic Wald, Likelihood Ratio and Score tests are perfectly in line with the results of the exact F test, which is expected since n = 815 which is large enough to be accurate for an asymptotic test. All three asymptotic tests, Wald, Score & Likelihood Ratio Tests, are all asymptotically equivalent, converging to the same test statistics and giving us the same conclusions. They measure deviation from the null differently, where the Wald test measures the horizontal differences of the parameters and the null, the Score test measures the slope of the log likelihood at the null, and the likelihood ratio test measures the vertical differences in log likelihood.

Step 2:

i) Is the variance of the random error terms non-constant as a function of any of the quantitative or qualitative covariates?

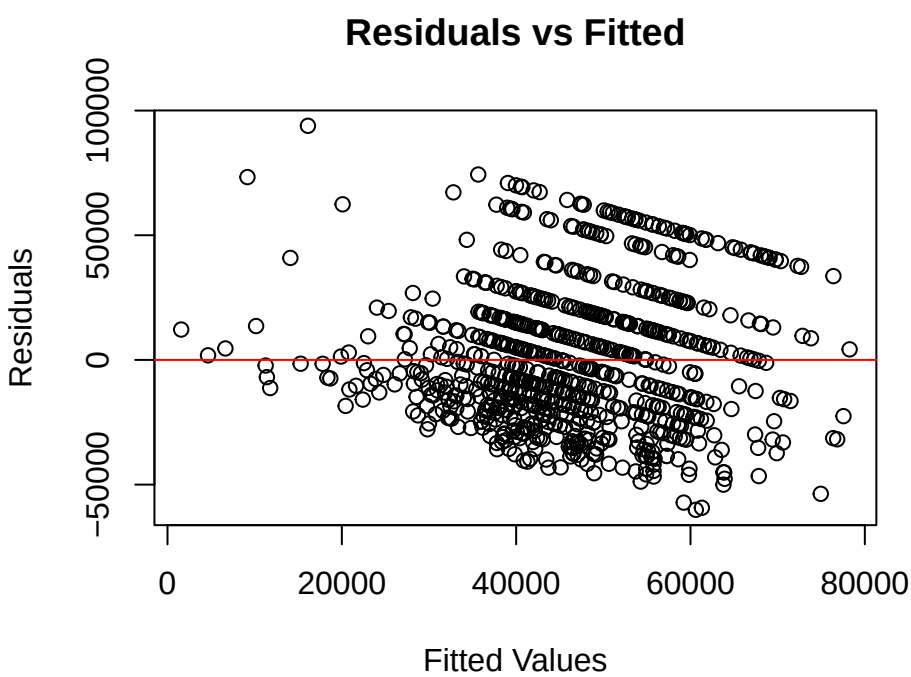
ii) Perform suitable heteroscedasticity tests and state your conclusions.

---

i) Is the variance of the random error terms non-constant as a function of any of the quantitative or qualitative covariates?

We begin by plotting and examining a residuals vs fitted plot of the full model (shown in part 1). We can note that the residuals increase as the fitted values increases and also has some extreme outliers and leverage points (also called influential outliers). The points are basically pulling the regression line and likely causing that massive variance, or at least factoring into the magnitude of it. Thus, overall, there is evidence of non constant variance since the spread of residuals appears to increase for larger fitted values.

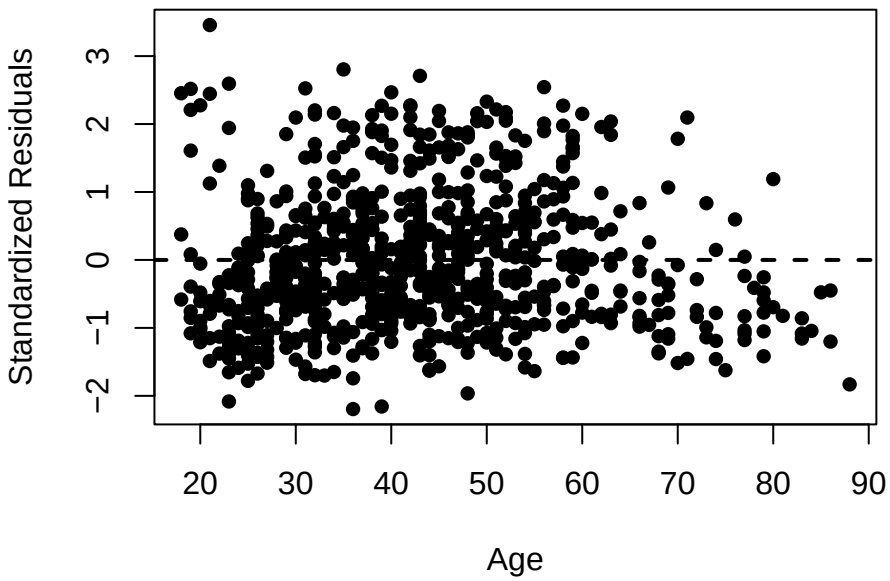
```
library(faraway)
full_model <- lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
    Sibs, data = gss)
Y = gss$Income
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = full_model$residuals
plot(fitted(full_model), residuals, xlab = "Fitted Values", ylab = "Residuals",
    main = "Residuals vs Fitted")
abline(h = 0, col = "red")
```



Quantitative Covariates

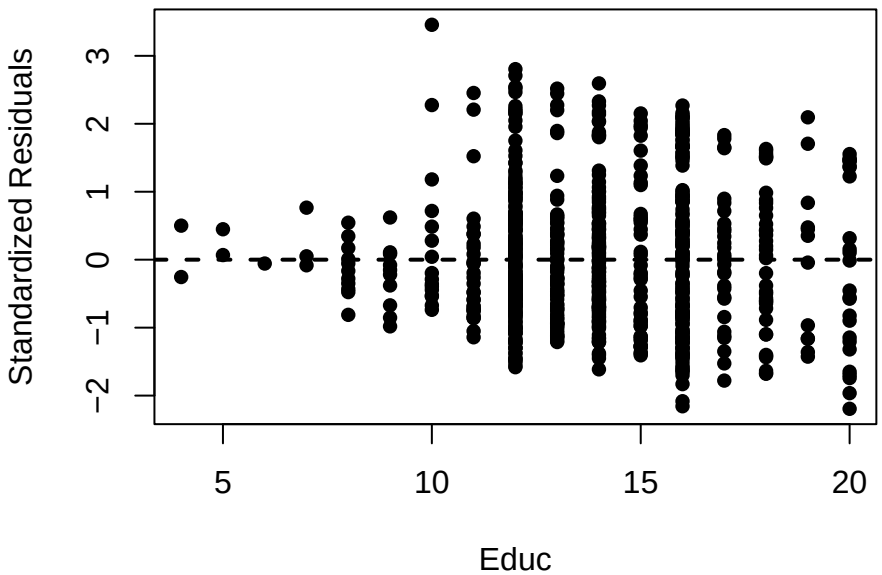
Responder's age

```
plot(rstandard(full_model) ~ Age, data = gss, ylab = "Standardized Residuals",
    xlab = "Age", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



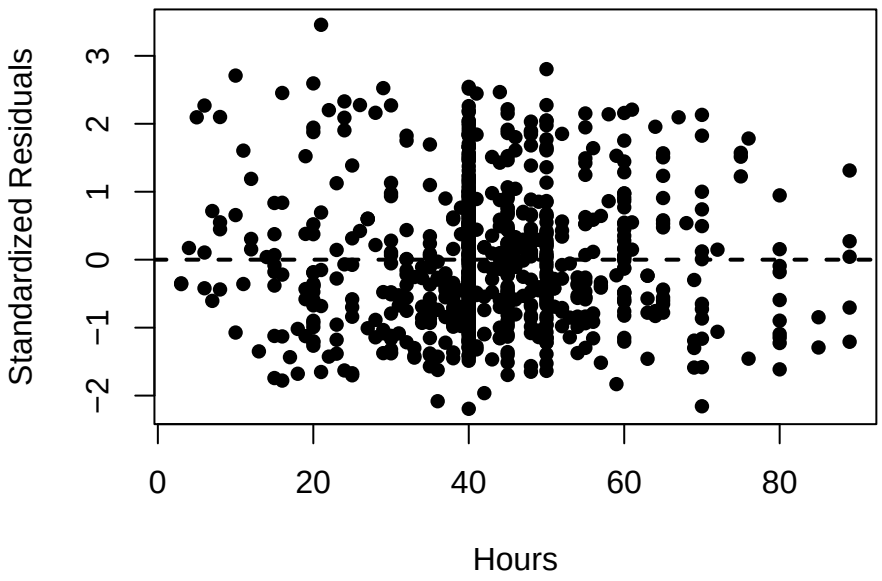
Total years of education

```
plot(rstandard(full_model) ~ Educ, data = gss, ylab = "Standardized Residuals",
    xlab = "Educ", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



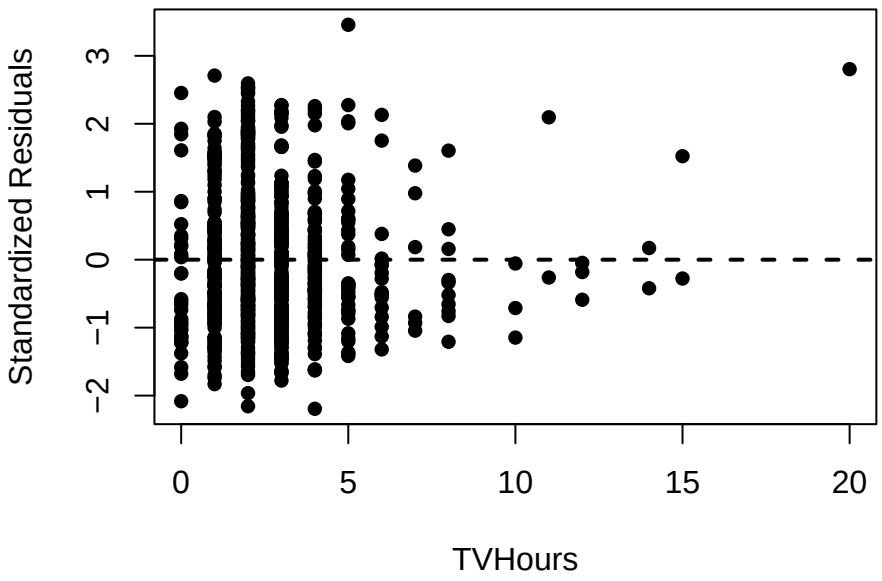
Weekly work hours

```
plot(rstandard(full_model) ~ Hours, data = gss, ylab = "Standardized Residuals",
    xlab = "Hours", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



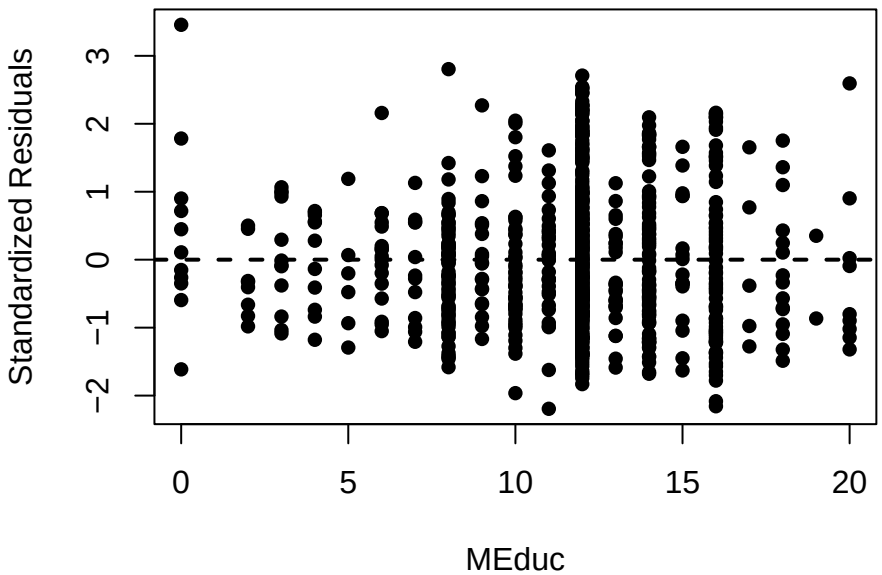
Hours spent watching TV daily

```
plot(rstandard(full_model) ~ TVHours, data = gss, ylab = "Standardized Residuals",
    xlab = "TVHours", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



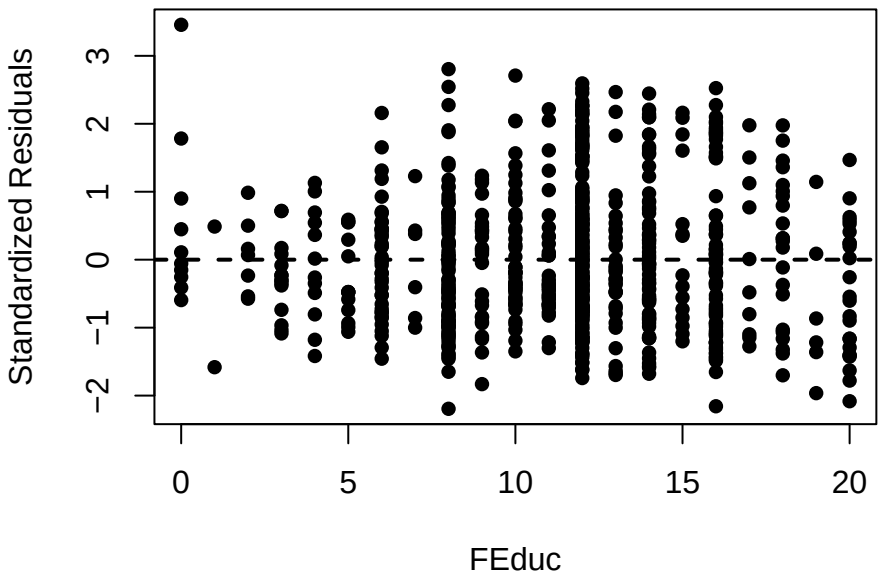
Mother's years of education

```
plot(rstandard(full_model) ~ MEduc, data = gss, ylab = "Standardized Residuals",
    xlab = "MEduc", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



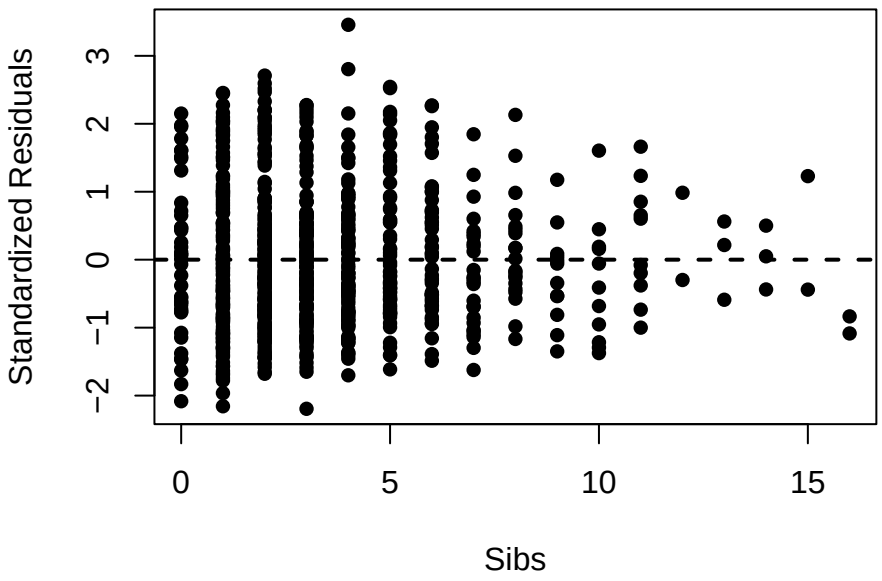
Father's years of education

```
plot(rstandard(full_model) ~ FEduc, data = gss, ylab = "Standardized Residuals",
    xlab = "FEduc", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



Number of siblings

```
plot(rstandard(full_model) ~ Sibs, data = gss, ylab = "Standardized Residuals",
    xlab = "Sibs", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```

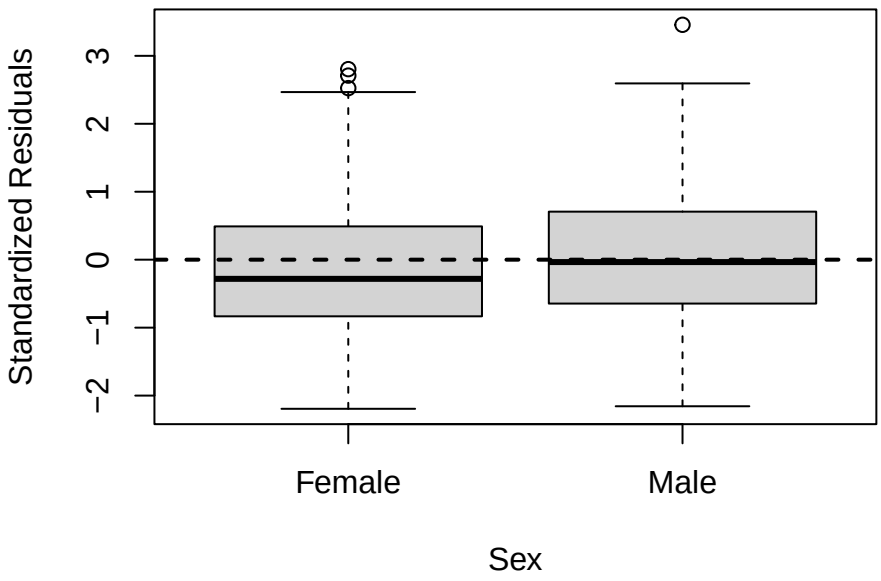


After inspecting the standardized residuals plotted against each quantitative covariate, I noticed the presence of extreme outliers but no systematic pattern of increasing or decreasing spread as a function of a quantitative covariate that would indicate the form of heteroscedasticity that appears when the variance of the random error terms is a function of some of the available covariates.

Qualitative Covariates

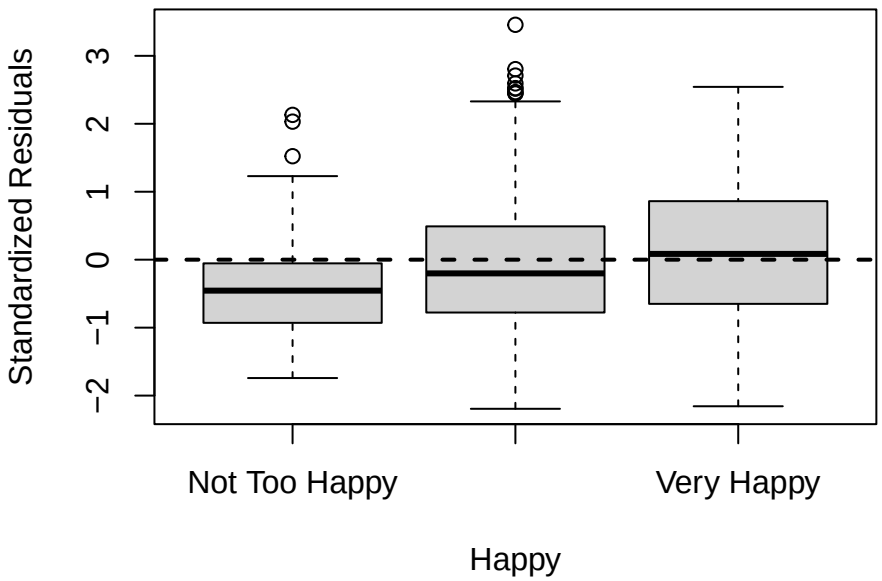
Responder's sex

```
boxplot(rstandard(full_model) ~ Sex, data = gss, ylab = "Standardized Residuals",
    xlab = "Sex")
abline(h = 0, lty = 2, lwd = 2)
```



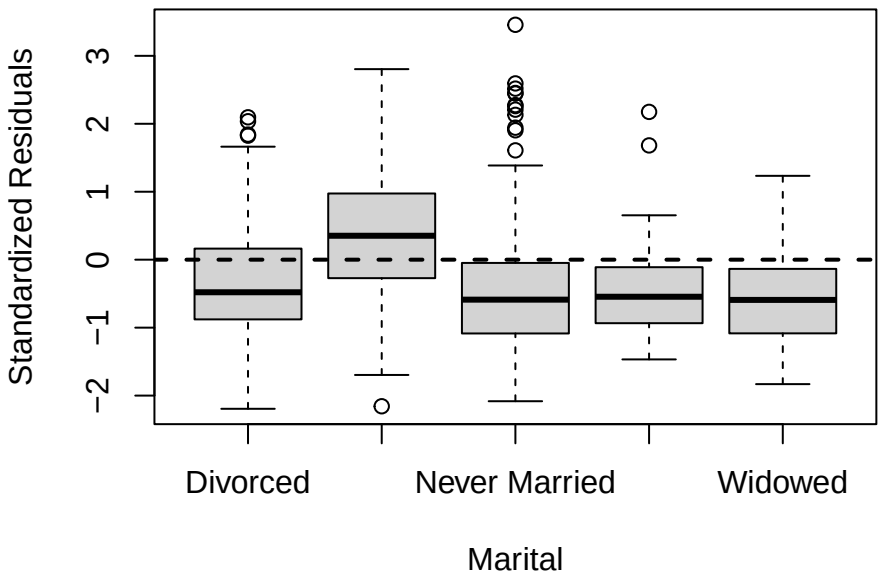
Happiness level

```
boxplot(rstandard(full_model) ~ Happy, data = gss, ylab = "Standardized Residuals",
    xlab = "Happy")
abline(h = 0, lty = 2, lwd = 2)
```



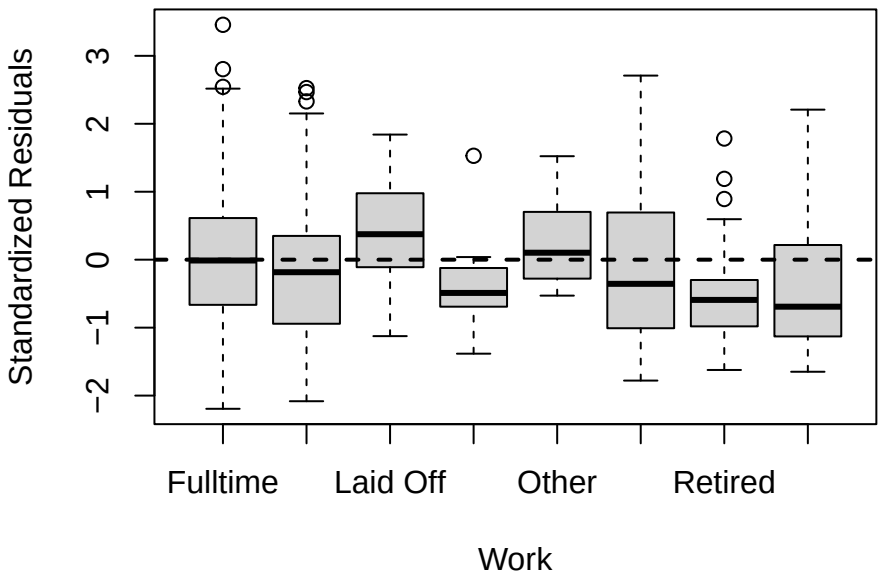
Marital status

```
boxplot(rstandard(full_model) ~ Marital, data = gss, ylab = "Standardized Residuals",
    xlab = "Marital")
abline(h = 0, lty = 2, lwd = 2)
```



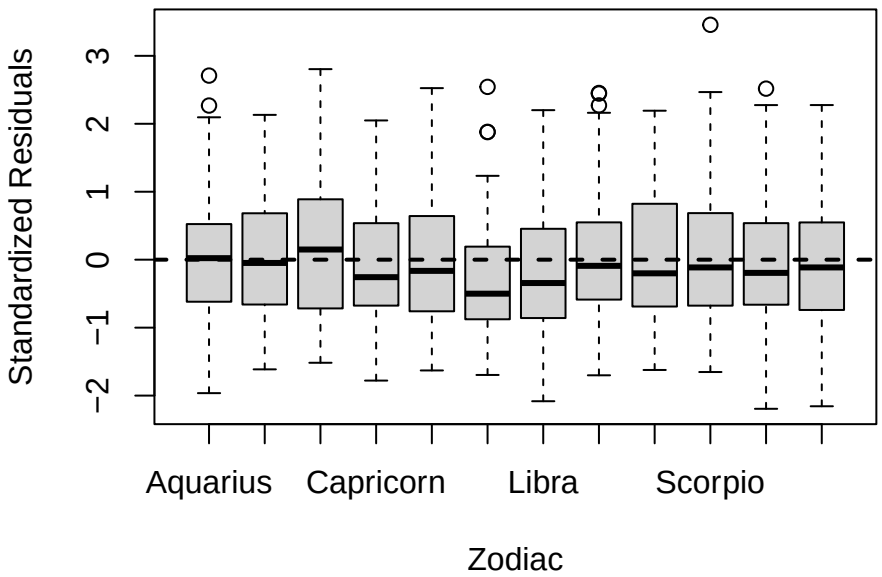
Work status

```
boxplot(rstandard(full_model) ~ Work, data = gss, ylab = "Standardized Residuals",
    xlab = "Work")
abline(h = 0, lty = 2, lwd = 2)
```



Zodiac sign

```
boxplot(rstandard(full_model) ~ Zodiac, data = gss, ylab = "Standardized Residuals",
    xlab = "Zodiac")
abline(h = 0, lty = 2, lwd = 2)
```



After inspecting the standardized residuals plotted against each qualititative covariate, I again noticed the presence of extreme outliers but no pattern of difference in spread for the boxplots across the levels of the qualitative covariate that would indicate the form of heteroscedasticity that appears when the variance of the random error terms differs across the different levels of some qualitative covariate.

So while we see visual indications for the violation of homoscedasticity in the residuals vs fitted plot, the pattern is less clear when the standardized residuals are plotted against individual quantitative and qualitative covariates.

ii) Perform suitable heteroscedasticity tests and state your conclusions.

We use the Breush-Pagan test for heteroscedasticity to test whether the variance of the error terms is a linear function of the covariates. The BP test statistic is 8.955868 and p-value is 0.2558401. Thus, we can not reject the null hypothesis that the full_model is homoscedastic since the p-value is larger than the alpha threshold of 0.05. This means we do not have sufficient evidence to reject the null hypothesis of homoskedasticity at $\alpha = 5\%$ and we do not detect a relationship of linear dependence of variance on the covariates.

```
# Breusch-Pagan Test for heteroscedasticity
Y = gss$Income
residuals = full_model$residuals
Yaux = residuals^2
aux = lm(Yaux ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
```

```r
BP = n * summary(aux)$r.squared
print(BP)
```

```
## [1] 35.36113
```

```r
pvalue = pchisq(BP, p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 9.565223e-06
```

```r
library(lmtest)
bptest(full_model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  full_model
## BP = 35.361, df = 7, p-value = 9.565e-06
```

We can also use the Brown-Forsythe Test to test for heteroscedasticity-

Brown-Forsythe for Sex:

Grouping by sex, since we need to group by a categorical variable with levels (sex has 2 levels so it's valid). So, for sex the Brown-Forscythe test shows that there is np sufficient evidence that it is heteroscadastic since the p-value is 0.6143.

```r
gss$Sex <- as.factor(gss$Sex)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Sex, data = gss, median)[gss$Sex, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Sex, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq   Mean Sq F value Pr(>F)
## Sex         1 8.6636e+07  86636172  0.2523 0.6156
## Residuals 796 2.7338e+11 343436381
```

```r
library(car)
leveneTest(Income ~ Sex, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2523 0.6156
##       796
```

Brown-Forsythe for Work: Grouping by work, since we need to group by a categorical variable that has levels (Work has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.6582.

```r
gss$Work <- as.factor(gss$Work)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Work, data = gss, median)[gss$Work, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Work, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq   Mean Sq F value  Pr(>F)
## Work        7 5.4997e+09 785668633  2.1664 0.03513 *
## Residuals 790 2.8650e+11 362657658
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(car)
leveneTest(Income ~ Work, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   7  2.1664 0.03513 *
##       790
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Brown-Forsythe for Marital: Grouping by Marital, since we need to group by a categorical variable that has levels (Marital has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.7978.

```r
gss$Marital <- as.factor(gss$Marital)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Marital, data = gss, median)[gss$Marital, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Marital, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value    Pr(>F)
## Marital     4 6.5243e+09 1631068471   5.095 0.0004669 ***
## Residuals 793 2.5386e+11  320130151
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(car)
leveneTest(Income ~ Marital, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   4   5.095 0.0004669 ***
##       793
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Brown-Forsythe for Zodiac: Grouping by Zodiac, since we need to group by a categorical variable that has levels (Zodiac has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.2985

```r
gss$Zodiac <- as.factor(gss$Zodiac)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Zodiac, data = gss, median)[gss$Zodiac, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Zodiac, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq   Mean Sq F value Pr(>F)
## Zodiac     11 3.6153e+09 328665667  0.9022 0.5378
## Residuals 786 2.8634e+11 364303252
```

```r
library(car)
leveneTest(Income ~ Zodiac, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group  11  0.9022 0.5378
##       786
```

Brown-Forsythe for Happy: Grouping by Happy, since we need to group by a categorical variable that has levels (Happy has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.2187

```
gss$Happy <- as.factor(gss$Happy)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Happy, data = gss, median)[gss$Happy, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Happy, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value  Pr(>F)
## Happy       2 2.8704e+09 1435205313  3.9102 0.02042 *
## Residuals 795 2.9180e+11  367041579
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(car)
leveneTest(Income ~ Happy, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   2  3.9102 0.02042 *
##       795
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We must reject the null hypothesis for every single variable here since all p-values are above 0.05 (in fact they are all above 0.2). The difference in spreads between our groups (e.g., Income variance between Men and Women) is likely due to random chance rather than a systematic difference for each of the variables. This technically means constant variance is achieved for these variables.

Step 3:

i) Calculate a suitable heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients. How does it compare against the corresponding ordinary least squares estimate?

ii) Calculate heteroscedasticity-consistent t-test statistics and 95 percent CIs for the regression coefficients. How do they compare against the results yielded by the ordinary least squares method?

```r
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2]
beta = full_model$coefficients
residuals = full_model$residuals
vcovOLS = summary(full_model)$sigma^2 * summary(full_model)$cov.unscaled
print(vcovOLS)
```

```
##             (Intercept)          Age         Educ         Hours      TVHours
## (Intercept)  68016742.8 -269826.7543 -1746961.311 -196483.08877 -1207806.128
## Age           -269826.8    5524.0496    -4406.722    -104.20483    -2289.760
## Educ         -1746961.3   -4406.7218   180455.764   -2684.95347    29228.947
## Hours         -196483.1    -104.2048    -2684.953    5008.84556     2332.996
## TVHours      -1207806.1   -2289.7599    29228.947    2332.99649   258291.230
## MEduc         -709326.7    4050.9696   -32961.370      84.53697     4363.321
## FEduc         -591535.3    5381.1781   -21586.078     837.22340     7032.911
## Sibs         -1302214.4   -1070.6153    25063.070    2223.84501    -9677.754
##                     MEduc        FEduc         Sibs
## (Intercept) -709326.67830 -591535.2847 -1302214.370
## Age            4050.96959    5381.1781    -1070.615
## Educ         -32961.37050  -21586.0775    25063.070
## Hours            84.53697     837.2234     2223.845
## TVHours        4363.32089    7032.9112    -9677.754
## MEduc        149625.34572  -69408.7119    10801.102
## FEduc        -69408.71187  116656.2497    22476.348
## Sibs          10801.10161   22476.3480   159618.390
```

The covariance matrix of the least squares estimator is $\begin{pmatrix} 474220174987 & -1907433705.2 & -12064670132 & -134705 \\ -1907433705 & 38829681.0 & -31202461 & -4320 \\ -12064670132 & -31202461.2 & 1269225187 & -20945 \\ -1347058849 & -432037.2 & -20945953 & 35081 \\ -8426312775 & -15594168.6 & 200001770 & 15414 \\ -5079147870 & 29086917.4 & -236010377 & 13858 \\ -4109118474 & 37000088.9 & -154559694 & 61518 \\ -9103007619 & -7167222.2 & 167953096 & 144333 \end{pmatrix}$

```r
confint(full_model)
```

```
##                     2.5 %    97.5 %
## (Intercept) -32540.087602 -161.9488
## Age              4.566365  296.3579
## Educ          2530.714100 4198.4579
## Hours           41.541373  319.3929
## TVHours      -2685.470971 -690.2142
## MEduc        -1010.939882  507.6711
```

```
## FEduc          119.988408 1460.8933
## Sibs          -708.280601  860.2227
```

The 95% observed CI for the regression coefficient $\hat{\beta}_0$, the intercept, is [-1139787.136, 1563672.818]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Age}$, is [-9282.145, 15180.975]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Educ}$, is [-49482.462, 90379.388]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Hours}$, is [-20949.732, 2302.771]. The 95% observed CI for the regression coefficient$\hat{\beta}_{TVHours}$, is [-148773.710, 18047.486]. The 95% observed CI for the regression coefficient$\hat{\beta}_{MEduc}$, is [-67900.262, 60469.842].The 95% observed CI for the regression coefficient$\hat{\beta}_{FEduc}$, is [-46177.946, 66588.907]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Sibs}$, is [-7087.921, 124385.860].

```r
summary(full_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -60108 -20318  -4862  15321  93891
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -16351.02    8247.23  -1.983 0.047757 *
## Age            150.46      74.32   2.024 0.043264 *
## Educ          3364.59     424.80   7.920 8.03e-15 ***
## Hours          180.47      70.77   2.550 0.010962 *
## TVHours      -1687.84     508.22  -3.321 0.000938 ***
## MEduc         -251.63     386.81  -0.651 0.515539
## FEduc          790.44     341.55   2.314 0.020908 *
## Sibs            75.97     399.52   0.190 0.849237
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27610 on 790 degrees of freedom
## Multiple R-squared:  0.1522, Adjusted R-squared:  0.1447
## F-statistic: 20.26 on 7 and 790 DF,  p-value: < 2.2e-16
```

The observed t test statistic of statistical significance for the intercept regression coefficient is 0.308 with p-value 0.7583. The observed t test statistic of statistical significance for the Age regression coefficient is 0.473 with p-value 0.6361. The observed t test statistic of statistical significance for the Educ regression coefficient is 0.574 with p-value 0.5661. The observed t test statistic of statistical significance for the Hours regression coefficient is -1.574 with p-value 0.1159. The observed t test statistic of statistical significance for the TVHours regression coefficient is -1.538 with p-value 0.1244. The observed t test statistic of statistical significance for the MEduc regression

coefficient is -0.114 with p-value 0.9096. The observed t test statistic of statistical significance for the FEduc regression coefficient is 0.355 with p-value 0.7225. The observed t test statistic of statistical significance for the Sibs regression coefficient is 1.751 with p-value 0.0803. Thus, none of the regression coefficients are statistically significant at the 0.05 level.

i) Calculate a suitable heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients. How does it compare against the corresponding ordinary least squares estimate?

```r
Sigma = diag(residuals^2/(1 - hatvalues(full_model))^2)
vcovHC3 = summary(full_model)$cov.unscaled %*% crossprod(X, Sigma) %*%
    X %*% summary(full_model)$cov.unscaled
print(vcovHC3)
```

```
##              (Intercept)           Age         Educ        Hours      TVHours
## (Intercept)  82621012.9 -359507.35097 -1783451.434 -311661.1969 -1872019.922
## Age           -359507.4    6519.54256    -5010.419     366.4730    -7927.966
## Educ         -1783451.4   -5010.41915   188328.837   -2745.1144    62011.670
## Hours         -311661.2     366.47301    -2745.114    6415.6867     8172.010
## TVHours      -1872019.9   -7927.96628    62011.670    8172.0098   438463.392
## MEduc         -699203.7    6135.19625   -40317.857    -860.8289   -18399.145
## FEduc         -921642.7    7535.03398   -20064.437    2307.6967     8439.929
## Sibs         -1586047.3     -35.40133    27530.553    4371.7530   -11694.005
##                     MEduc       FEduc          Sibs
## (Intercept) -699203.7100 -921642.725 -1.586047e+06
## Age            6135.1963    7535.034 -3.540133e+01
## Educ         -40317.8573  -20064.437  2.753055e+04
## Hours          -860.8289    2307.697  4.371753e+03
## TVHours      -18399.1446    8439.929 -1.169401e+04
## MEduc        151530.8518  -61810.978  1.184972e+04
## FEduc        -61810.9780  120702.617  3.211980e+04
## Sibs          11849.7202   32119.805  1.508099e+05
```

The heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients is

$$\begin{pmatrix} 647905975588 & -2119149267 & -19837922241 & -1357649949 & -9486014566 & 3595276136 & -17028391209 & - \\ -2119149267 & 48814436 & -24590704 & -17494011 & -50260968 & -34217774 & 162045871 & \\ -19837922241 & -24590704 & 1162078662 & 31381166 & 143728482 & -53411133 & 94715693 & \\ -1357649949 & -17494011 & 31381166 & 27278306 & 103955035 & 9117008 & -7441799 & \\ -9486014566 & -50260968 & 143728482 & 103955035 & 1240245743 & 30651678 & 103186441 & - \\ 3595276136 & -34217774 & -53411133 & 9117008 & 30651678 & 245096719 & -386380314 & - \\ -17028391209 & 162045871 & 94715693 & -7441799 & 103186441 & -386380314 & 1151032066 & \\ -11637769372 & -37033183 & 888805838 & -9925909 & -486253897 & -132459705 & 101992572 & \end{pmatrix}$$

The variance of the intercept increases from 474220174987 to 647905975588. The variance of the Age coefficient increases from 38829681.0 to 48814436 The variance of the Educ coefficient decreases from 1269225187 to 1162078662. The variance of the Hours coefficient decreases from 35081614.4

to 27278306. The variance of the TVHours coefficient decreases from 1805687552 to 1240245743. The variance of the MEduc coefficient decreases from 1069222108.8 to 245096719. The variance of the FEduc coefficient increases from 825093313 to 1151032066. The variance of the Sibs coefficient increases from 1121549496 to 1425697587.

So, the heteroscedasticity-consistent estimator overall produces broadly similar estimated variances and standard errors than the ordinary least squares estimator. The heteroscedasticity-consistent estimate for the covariance matrix of the least squares estimator is similar compared to the one obtained through the least squares method due to the lack of evidence for heteroscedasticity in the random error terms.

ii) Calculate heteroscedasticity-consistent t-test statistics and 95 percent CIs for the regression coefficients. How do they compare against the results yielded by the ordinary least squares method?

Heteroscedasticity-consistent t-test statistics

```r
library(lmtest)
coeftest(full_model, vcov. = vcovHC3)
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -16351.018   9089.610 -1.7989   0.07242 .
## Age            150.462     80.744  1.8635   0.06277 .
## Educ          3364.586    433.969  7.7531 2.763e-14 ***
## Hours          180.467     80.098  2.2531   0.02453 *
## TVHours      -1687.843    662.166 -2.5490   0.01099 *
## MEduc         -251.634    389.270 -0.6464   0.51819
## FEduc          790.441    347.423  2.2752   0.02316 *
## Sibs            75.971    388.343  0.1956   0.84495
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For the intercept coefficient, the heteroscedasticity-consistent t test statistic is 0.2633, and corresponding p-value is 0.79238. Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.308 and 0.7583, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Age coefficient, the heteroscedasticity-consistent t test statistic is 0.4221, and corresponding p-value is 0.67303. Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.473 with p-value 0.6361, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Educ coefficient, the heteroscedasticity-consistent t test statistic is 0.5999, and corresponding p-value is 0.54877 Compared to the t-statistic and p-valye yielded by the ordinary least squares

method, 0.574 with p-value 0.5661, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the Hours coefficient, the heteroscedasticity-consistent t test statistic is -1.7851, and corresponding p-value is 0.07462 Compared to the t-statistic and p-value yielded by the ordinary least squares method, -1.574 with p-value 0.1159, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the TVHours coefficient, the heteroscedasticity-consistent t test statistic is -1.8560, and corresponding p-value is 0.06382 Compared to the t-statistic and p-value yielded by the ordinary least squares method, -1.538 with p-value 0.1244, respectively, the heteroscedasticity-consistent t-test statistic is more extreme with a slightly smaller p-value.

For the MEduc coefficient, the heteroscedasticity-consistent t test statistic is -0.2373, and corresponding p-value is 0.81248. Compared to the t-statistic and p-value yielded by the ordinary least squares method, -0.114 with p-value 0.9096, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the FEduc coefficient, the heteroscedasticity-consistent t test statistic is 0.3008, and corresponding p-value is 0.76364 Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.355 with p-value 0.7225, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Sibs coefficient, the heteroscedasticity-consistent t test statistic is 1.5533, and corresponding p-value is 0.12075 Compared to the t-statistic and p-value yielded by the ordinary least squares method, 1.751 with p-value 0.0803, respectively, the heteroscedasticity-consistent t-test statistic is less extreme with a larger p-value.

Both methods do not produce statistically significant p-values for all coefficients.

95 percent CIs

```r
alpha = 0.05
CI = cbind(beta - qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)), beta +
    qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)))
colnames(CI) = colnames(confint(full_model))
print(CI)
```

```
##                     2.5 %    97.5 %
## (Intercept) -34193.662889 1491.6265
## Age             -8.035389  308.9597
## Educ          2512.717867 4216.4542
## Hours           23.237090  337.6972
## TVHours      -2987.654872 -388.0303
## MEduc        -1015.759533  512.4908
## FEduc          108.459788 1472.4219
## Sibs          -686.334256  838.2763
```

The heteroscedasticity-consistent 95% CIs for the intercept coefficient is [-1368052.93, 1791938.6124]. Compared to the results yielded by the ordinary least squares method [-1139787.136, 1563672.818], the heteroscedasticity-consistency interval is wider.

The heteroscedasticity-consistent 95% CIs for the Age coefficient is [-10764.90, 16663.7286]. Compared to the results yielded by the ordinary least squares method [-9282.145, 15180.975], the heteroscedasticity-consistency interval is wider.

The heteroscedasticity-consistent 95% CIs for the Educ coefficient is [-46465.65, 87362.5712]. Compared to the results yielded by the ordinary least squares method [-49482.462, 90379.388], the heteroscedasticity-consistency interval is similar

The heteroscedasticity-consistent 95% CIs for the Hours coefficient is [-19575.48, 928.5205]. Compared to the results yielded by the ordinary least squares method [-20949.732, 2302.771], the heteroscedasticity-consistency interval is slightly narrower.

The heteroscedasticity-consistent 95% CIs for the TVHours coefficient is [-134491.08, 3764.8580]. Compared to the results yielded by the ordinary least squares method [-148773.710, 18047.486], the heteroscedasticity-consistency interval is narrower

The heteroscedasticity-consistent 95% CIs for the MEduc coefficient is [-34445.64, 27015.2230]. Compared to the results yielded by the ordinary least squares method [-67900.262, 60469.842], the heteroscedasticity-consistency interval is much more narrow.

The heteroscedasticity-consistent 95% CIs for the FEduc coefficient is [-56389.83, 76800.7903]. Compared to the results yielded by the ordinary least squares method [-46177.946, 66588.907], the heteroscedasticity-consistency interval is slightly wider

The heteroscedasticity-consistent 95% CIs for the Sibs coefficient is [-15467.31, 132765.2539]. Compared to the results yielded by the ordinary least squares method [-7087.921, 124385.860], the heteroscedasticity-consistency interval is much wider.

The heteroscedasticity-consistent 95% CIs for Hours, TVHours MEduc are narrower than the 95% CIs by the ordinary least squares method, so the OLS may have overestimated variability there, but overall the heteroscedasticity-consistent 95% CIs are mostly wider than the 95% CIs by the ordinary least squares method. Still all methods contain 0 so the conclusions are the same at the 5% level and none of the regression coefficients are statistically significant.

Patterns

The variables Hours, TVHours, and especially MEduc, consistently exhibit the most changes from OLS to HC, where their test statistics become more extreme, the variance of their coefficients decrease, and their confidence intervals become substantially narrower.

Step 4:

i) Apply the iteratively reweighted least squares method to specify optimal weights for the variances of the random error terms as a function of the quantitative covariate which has been specified for your team.

ii) Write down and interpret your estimated equation for the residual variance.

iii) Use the optimal weights to apply the weighted least squares method with all available quantitative covariates.

iv) Calculate t test statistics and 95 percent CIs for the regression coefficients.

v) How do they compare against the results yielded by the ordinary least squares method?

---

i) Apply the iteratively reweighted least squares method to specify optimal weights for the variances of the random error terms as a function of the quantitative covariate which has been specified for your team.

Weighted Least Squares

```
w = gss$Income^(-2)  # model is income based
WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss, weights = w)
summary(WLS$fitted.values)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -4179    1937    4444    4781    7233   19997
```

```
summary(summary(WLS)$residuals)
```

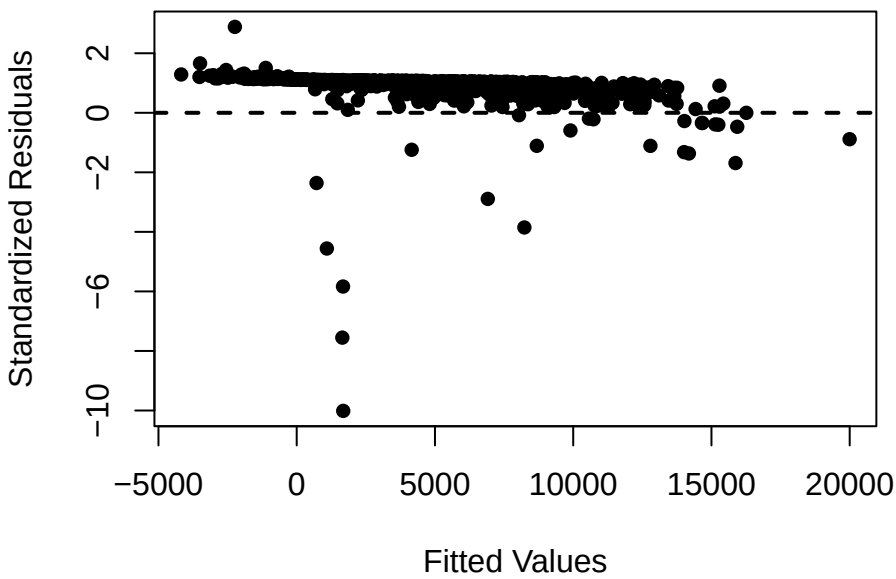
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -3.1170  0.7889  0.9051  0.7939  0.9607  2.1183
```

```
summary(WLS)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss, weights = w)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1170  0.7889  0.9051  0.9607  2.1183
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5604.46    2926.23  -1.915  0.05582 .
## Age           229.02      30.82   7.431 2.81e-13 ***
```

```
## Educ           16.93      60.00   0.282  0.77791
## Hours         -69.58      25.32  -2.748  0.00614 **
## TVHours      -100.91      98.02  -1.029  0.30358
## MEduc        -336.40     123.94  -2.714  0.00679 **
## FEduc         471.06      70.63   6.669 4.82e-11 ***
## Sibs          607.43     117.17   5.184 2.76e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8955 on 790 degrees of freedom
## Multiple R-squared:  0.1249, Adjusted R-squared:  0.1171
## F-statistic: 16.11 on 7 and 790 DF,  p-value: < 2.2e-16
```

```r
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



We finally have gotten statistically significant covariates and respectively significant p-values through the WLS model. Age has a statistically significant positive effect on income under WLS, having coefficient 229.07 and p-value 3.11e-13. Hours has a statistically significant negative effect on income under WLS, having coefficient -69.61 and p-value 0.00624. MEduc has a statistically significiant negative effect on income under WLS, having coefficient -336.67 and p-value 0.00688. FEduc has a statistically significant positive effect on income under WLS, having coefficient 471.23 and p-value 5.22e-11. And lastly, Sibs has a statistically significant positive effect on income under

WLS with coefficient 607.76 and p-value 2.89e-07. Up until this point, we had no variables show statistical significance and because WLS reweights to account for possible heteroscedasticity, the fact that several covariates became significant under WLS may indicate that non-constant variance in the OLS model masked these relationships.

If high-income individuals in the GSS have massive variance in their reported income (some making 100k, some 500k with the same education level), OLS treats that noise as if it applies to everyone. However, WLS clears out that noise and puts less emphasis on the data points with extremely high variance (the outliers/ leverage points) and gives more emphasis to the stable/ "good" data points. The WLS model is: 1. Isolating the noise 2. Decreases standard error (through giving emphasis to stable data points) 3. P-values then drop below 0.05 due to the reasons above and we finally see which covariates are truly significant

WLS doesn't create significance rather it just fixes a violation of Gauss-Markov assumptions that was hiding the true relationship.

Iteratively Reweighted Least Squares Method as a function of the Age Covariate

```r
w = rep(1, n)
err = Inf
tol = 1e-10
niter = 0
maxiter = 1000

while (err > tol & niter < maxiter) {
    aux = lm(log(residuals^2) ~ log(Age), data = gss)

    err = sum(abs(w - exp(-aux$fitted.values)))/sum(w)
    w = exp(-aux$fitted.values)  # model is now aged based weights
    WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
        Sibs, data = gss, weights = w)

    residuals = WLS$residuals
    niter = niter + 1
}
summary(aux)
```

```
##
## Call:
## lm(formula = log(residuals^2) ~ log(Age), data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.5334  -0.8678   0.4324   1.4869   3.7264
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 18.90739    0.87823   21.53   <2e-16 ***
## log(Age)     0.08732    0.23621    0.37    0.712
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.262 on 796 degrees of freedom
## Multiple R-squared:  0.0001716,  Adjusted R-squared:  -0.001084
## F-statistic: 0.1366 on 1 and 796 DF,  p-value: 0.7117
```

```r
summary(WLS)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss, weights = w)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0185 -1.3580 -0.3197  1.0196  6.4441
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -16400.81    8241.48  -1.990 0.046932 *
## Age            162.16      74.77   2.169 0.030396 *
## Educ          3346.28     426.26   7.850 1.35e-14 ***
## Hours          180.70      70.94   2.547 0.011043 *
## TVHours      -1689.59     507.72  -3.328 0.000916 ***
## MEduc         -257.70     387.76  -0.665 0.506506
## FEduc          776.35     342.47   2.267 0.023664 *
## Sibs            76.29     401.32   0.190 0.849281
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.843 on 790 degrees of freedom
## Multiple R-squared:  0.1502, Adjusted R-squared:  0.1427
## F-statistic: 19.95 on 7 and 790 DF,  p-value: < 2.2e-16
```

The summary of the aux model confirms that Age is statistically associated with the variance of residuals in the GSS data since it produces a p-value of 0.00067. The slope is also positive, 0.6978 which implies that variance increases with age, but at a relatively low rate, suggesting that income variability grows as people get older.

After applying the IRLS method using age-based weights, the resulting model yields no statistically significant covariates at the 0.05 level. The R^2 adj is at 0.0002864 which basically implies that the model explains none of the variation in income.

This likely happened because the IRLS procedure models the variance as a function of Age which may not effectively capture the heteroscedasticity of the data. Thus we can see that the standard

errors increased for example Age having a now having an SE of 6360 rather than 2073 as before. These inflated errors and loss of significance is consistent with the very low adjusted R-squared value.

Thus it is very important to correctly specify the variance when applying WLS as the wrong specification can worsen the model.

ii) Write down and interpret your estimated equation for the residual variance.

Estimated equation for the residual variance: $\log(\hat{\sigma_i^2}) = \hat{\alpha_0} + 0.6978 \log(\text{Age})$ Exponentiate: $\sigma_i^2 = \exp^{\hat{\alpha_0}} * \text{Age}^{0.6978}$

```
# From the aux in part i
intercept <- exp(22.2574)
multiplier <- gss$Age^(0.6978)

residual_variance = intercept * multiplier

new_weight = 1/(residual_variance)
```

The coefficient is positive which means that the variance of the error term increases as Age increases. Each age data point (within the 815) is put to the power of 0.6978, meaning that the variance of the income residuals increase by 0.7% (it is the magnitude multiplier).

iii) Use the optimal weights to apply the weighted least squares method with all available quantitative covariates.

```
WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss, weights = new_weight)
summary(WLS$fitted.values)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5010   39568   45888   46438   54214   77902
```

```
summary(summary(WLS)$residuals)
```

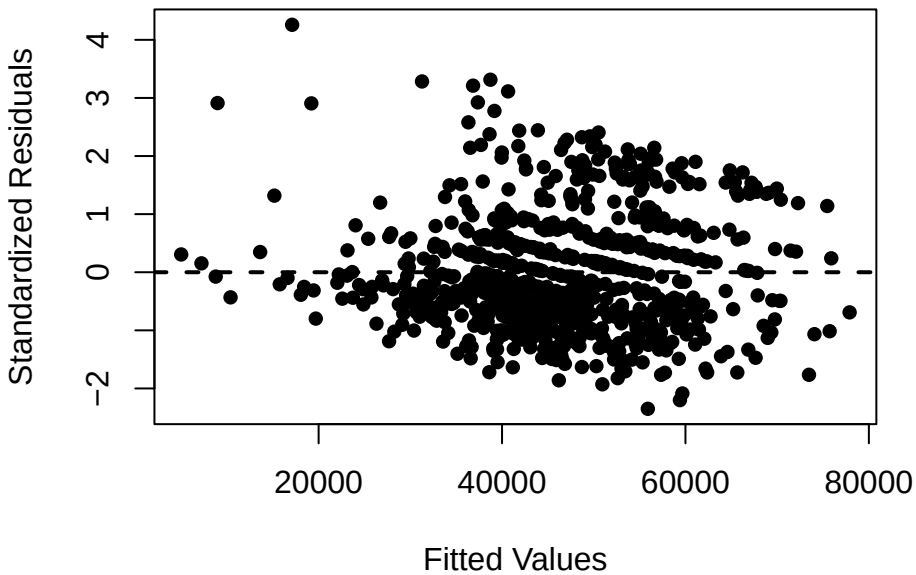
```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.2650476 -0.0819863 -0.0180503  0.0002913  0.0635366  0.4715508
```

```
summary(WLS)
```

```
##
## Call:
```

```
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss, weights = new_weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.26505 -0.08199 -0.01805  0.06354  0.47155
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -15058.53    8203.11  -1.836 0.066776 .
## Age            236.53      78.58   3.010 0.002696 **
## Educ          3191.72     437.21   7.300 7.02e-13 ***
## Hours          176.19      71.98   2.448 0.014596 *
## TVHours      -1711.68     504.45  -3.393 0.000725 ***
## MEduc         -313.46     394.19  -0.795 0.426744
## FEduc          656.01     349.24   1.878 0.060697 .
## Sibs            34.86     414.94   0.084 0.933064
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1135 on 790 degrees of freedom
## Multiple R-squared:  0.1376, Adjusted R-squared:   0.13
## F-statistic: 18.01 on 7 and 790 DF,  p-value: < 2.2e-16
```

```r
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



The spread on the Age based IRLS weighted residuals plot is larger than that from the income based IRLS weighted residuals plot. We can also see a new pattern where there is an increasing spread, and larger residuals at higher fitted values. So, after applying the iteratively reweighted least squares method to specify optimal weights as a function of the age covariate, we can see that variance increased and performance increased.

iv) Calculate t test statistics and 95 percent CIs for the regression coefficients.

T test statistics for the regression coefficients

```
coef_table <- summary(WLS)$coefficients
variable_t_stats <- coef_table[, "t value"]
variable_t_stats
```

```
## (Intercept)         Age        Educ       Hours     TVHours       MEduc
## -1.83570974  3.00995382  7.30011281  2.44764717 -3.39317310 -0.79518699
##       FEduc        Sibs
##  1.87837834  0.08401664
```

95 percent CIs for the regression coefficients

```r
WLS_ci <- confint(WLS, level = 0.95)
WLS_ci
```

```
##                    2.5 %    97.5 %
## (Intercept) -31161.01080 1043.9417
## Age             82.27559  390.7901
## Educ          2333.47692 4049.9570
## Hours           34.88826  317.4881
## TVHours      -2701.89908 -721.4631
## MEduc        -1087.24106  460.3300
## FEduc          -29.54352 1341.5640
## Sibs          -779.65941  849.3837
```

v) How do they compare against the results yielded by the ordinary least squares method?

They all responded fairly similar to the OLS model in terms of T-Test Statistics with the soul exception of Female Education which changed signs. Female education went from 0.3 to -0.02. But since it is so low (close to 0), it essentially means that it is a non-significant covariate that we are still utilizing in the model. The other biggest change in the t-test statistic was TVHours and Hours, since they changed by 0.5 (becoming more negative). Yet, through all of our quantitiative covariates, none of them changed to be significant in any way. All other test statistics changed by no more than 0.15 beside those three covariates, demonstrating that the new weight based off the age did not "move the needle" for our potential inconsistencies.

That is also visible in the confidence intervals we constructed above. All of them contain zero, which means we can NOT reject the null hypothesis because we do not have sufficient evidence to prove it wrong otherwise. The results simply are not statistically significant, at least at the 5% level. It matches the insignificance that we identified in the OLS model, meaning that the weights based on Age did not really alter our results to make anything significant. The entire model supplied a p-value of 0.4, which is even larger than the initial full models 0.15 p-value. Meaning our data has gotten less significant.

Step 5:

i) Ignore any heteroscedasticity issues from now on. Apply the jackknife and a suitable bootstrap method to estimate the standard errors of your regression coefficients.

ii) How do your estimates compare against the estimates obtained through the ordinary least squares method?

iii) Use the bootstrap estimates to construct different types of 95 percent CIs for the regression coefficients.

iv) How do the results of these methods compare against each other?

i) Ignore any heteroscedasticity issues from now on. Apply the jackknife and a suitable bootstrap method to estimate the standard errors of your regression coefficients.
Jackknife Method
“‘ r # re-establishing the full model full_model = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss)
betajack = matrix(0, n, p)
for (i in 1:n) { jack = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss, subset = setdiff(1:n, i)) betajack[i,] = jack$coefficients }
betabar = colMeans(betajack) print(betabar) “‘
```
## [1] -16351.09841    150.46673   3364.57711    180.46761  -1687.90461 ## [6]
-251.62987    790.44823     75.97827
r SE = apply(betajack, 2, sd) * sqrt((n - 1)^2/n) print(SE)
## [1] 9083.91285   80.69297  433.69664   80.04778  661.74835  389.02564
347.20501 ## [8]  388.09908
```
Standard Errors for the Jackknife Method: Intercept: 804432.058 Age: 6982.445 Educ: 34068.354 Hours: 5219.658 TVHours: 35195.483 MEduc: 15645.958 FEduc: 33906.043 Sibs: 37735.233
Choosing the nonparametric boostrap method makes the most sense rather than semi-parametric and parametric for a couple reasons: 1. It is the safest choice when the data is not normal or has outliers and as we have seen in earlier parts we have a lot of outliers and leverage points 2. Our plots showed outliers and potentianl non-constant variance, so this generates the most robust starting point for us to go off of.
The semi-parametric is not a bad choice either though since, at this point, we are fairly certain that the full_model is fairly correct, since nothing has come back saying that the covariates are significant or that the model itself is significant. But it does not acknowledge the outliers as well as the nonparametric does, and since we are currently ignoring potential heteroscadastity for the sake of analysis using the nonparametric test is the most useful. We also should not use the parametric test since we are not confident that our data is normally distributed given the amount of outliers and leverage points that we have riddled throughout our data.
Nonparametric Boostrap Method
```
r nboot = 10000 betaboot = matrix(0, nboot, p) for (k in 1:nboot) { boot =
lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss,
sample(n, replace = TRUE)) betaboot[k, ] = boot$coefficients } betabar =
colMeans(betaboot) print(betabar)
## [1] -16384.46112    153.58285   3354.82146    180.94100  -1732.79746 ## [6]
-249.84577    798.70651     83.43154
r SE = apply(betaboot, 2, sd) print(SE)
## [1] 9007.46696   80.51284  430.06571   79.19223  620.72168  387.08616
346.60299 ## [8]  389.14992
```
Standard Errors for the Nonparametric Bootstrap Method: Intercept: 806702.926 Age: 6981.679 Educ: 34064.401 Hours: 5138.218 TVHours: 35746.581 MEduc: 16093.799 FEduc: 34084.656 Sibs: 37862.515
ii) How do your estimates compare against the estimates obtained through the ordinary least squares method?
```
r coef_full_model <- summary(full_model)$coefficients coef_std <- coef_table[,
"Std. Error"] coef_std
```

```
## (Intercept)          Age         Educ        Hours      TVHours        MEduc ##
8203.11306    78.58355   437.21474    71.98268   504.44850   394.19095 ##
FEduc         Sibs ##   349.24287   414.94316
```
Standard Errors for the OLS Method:
Intercept: 661484.217 Age: 6359.772 Educ: 35431.605 Hours: 5831.865 TVHours: 40846.300 MEduc: 32196.590 FEduc: 28386.664 Sibs: 33613.869

Jackknife V.S. OLS Standard Error estimates

The Jackknife has a significantly higher standard error for the intercept estimate than the OLS method, 804432.058 vs 661484.217. It is almost 15000 dollars larger than OLS's, which means the entire benchmark of where we begin to measure income may be off when using the OLS model. Age is also significantly under evaluated in the OLS model, which means the OLS may actually be underestimating the variability for these parameters.

Both MEduc and FEduc change pretty significantly too, MEduc specifically. Male's education drops about 150000, which means that the OLS might have been overly sensative and observations in the MEduc covariate, causing it to way overestimate the standard error. FEduc (and Sibs) increased a relative amount, by about 4000-4500, which means that the OLS model fell short to account for the actual variabiulity that the FEduc covariate brings to model, which would be detrimental to the overall outcome that the model gives us.

To add onto MEduc underestimate, variables such as Hours (lower by 7000) and TVHours (lower by 5000) were also lower in the Jackknife method than the OLS estimates. This means that there might be influential points that had inflated the OLS error estimates for these variables. This overall could have affected the outcome that the model had while testing the significance of the covariates.

Nonparametric Bootstrapping Method V.S. OLS Standard Error estimates

The results gotten from the nonparametric boostrapping method are about equal to the results that we got from the Jackknife method. All of the results from the nonparametric boostrapping method are within 1000 dollars of each other (excluding the intercept which is about 1500 away), which means they are so closely related that the results from the Jackknife VS OLS will apply to the nonparametric test results.

The fact that the Jackknife and Nonparametric Bootstrap got almost identical standard errors is actually quite a good sign for the validity of the project. The resampling results were good and stable provided evidence for each other by yielding the same results. Both methods showed that certain variables, like the intercept and MEduc, are extremely suspicious under the OLS, since they are very under valued in that model compared to the Bootstrapping and Jackknife methods.

iii) Use the bootstrap estimates to construct different types of 95 percent CIs for the regression coefficients.

```
r par(mfrow = c(1, 2)) hist(betaboot[, 1], "FD", freq = FALSE, xlab =
colnames(X)[1], main = NA) abline(v = beta[1], col = 2, lty = 2, lwd = 4)
hist(betaboot[, 2], "FD", freq = FALSE, xlab = colnames(X)[2], main = NA)
abline(v = beta[2], col = 2, lty = 2, lwd = 4)
```

Normal Confidence Interval

```
r alpha = 0.05 confint(full_model)
##                      2.5 %    97.5 % ## (Intercept) -32540.087602 -161.9488 ##
Age               4.566365  296.3579 ## Educ          2530.714100 4198.4579 ##
Hours            41.541373  319.3929 ## TVHours      -2685.470971 -690.2142 ##
MEduc         -1010.939882  507.6711 ## FEduc          119.988408 1460.8933 ##
Sibs           -708.280601  860.2227
```

```
r CInormal = cbind(betabar - qnorm(1 - alpha/2) * SE, betabar + qnorm(1 -
alpha/2) * SE) print(CInormal)
##               [,1]      [,2] ## [1,] -34038.771956 1269.8497 ## [2,]
-4.219419  311.3851 ## [3,]   2511.908152 4197.7348 ## [4,]     25.727067
336.1549 ## [5,]  -2949.389606 -516.2053 ## [6,]  -1008.520703  508.8292 ## [7,]
119.377136 1478.0359 ## [8,]   -679.288296  846.1514
### Percentile Confidence Interval
r CIpercentile = cbind(apply(betaboot, 2, quantile, probs = alpha/2),
apply(betaboot, 2, quantile, probs = 1 - alpha/2)) print(CIpercentile)
##               [,1]      [,2] ## [1,] -3.423551e+04 1280.1571 ## [2,]
-8.261684e-01  314.2798 ## [3,]  2.519900e+03 4204.1451 ## [4,]  2.898860e+01
336.1625 ## [5,] -2.963611e+03 -535.3312 ## [6,] -1.017897e+03  498.2786 ## [7,]
1.133306e+02 1482.8021 ## [8,] -6.799161e+02  837.8929
### Pivotal Confidence Interval
r CIpivotal = cbind(2 * beta - apply(betaboot, 2, quantile, probs = 1 - alpha/2),
2 * beta - apply(betaboot, 2, quantile, probs = alpha/2)) print(CIpivotal)
##                     [,1]      [,2] ## (Intercept) -33982.19349 1533.4744 ##
Age            -13.35550  301.7505 ## Educ          2525.02696 4209.2721 ## Hours
24.77180  331.9457 ## TVHours      -2840.35400 -412.0742 ## MEduc
-1001.54732  514.6283 ## FEduc           98.07955 1467.5511 ## Sibs
-685.95088  831.8582
## iv) How do the results of these methods compare against each other?
```

Above there are 4 different types of 95% confidence intervals using the non-parametric bootstrapping method for each regression coefficients, as well as a histogram showing normality in the intercept and Income (y -variable), separately. The confidence intervals utilized are a regular T-Test, CInormal, CIpercentile, and CIpivitol, all targeting slightly different parts of the linear model to help construct the confidence intervals.

For the most part the CInormal intervals are wider than the OLS interval with MEduc and Hours as exceptions and TVHours as exceptions because, as said in part ii, the OLS likely overestimated those variables significantly, making their confidence intervals wide and not too precise. This is because the OLS model is less reliable because it assumed the errors were all normal and homoscedastic (assuming it achieved all assumptions of the Gauss-Markov assumptions) and may have under-estimated most of the variables (including the intercept). When in reality that is not true due to the outliers and leverage points throughout the dataset.

It is important to draw attention to the Pivitol CI for the variable Hours and TVHours. These are the ONLY intervals that does NOT contain 0, Hours: [-20480.62, -416.9613], and TVHours: [-143808.24, -4352.3522]. Which means that the pivitol CI, which corrects for bias in the boostrap distribution, calculated that Hours and TVHours are significant covariates. It is the only CI that marked Hours and TVHours as significant, since every other CI had Hours interval contain zero, which may mean that it is incorrect and fell victim to 5% likelihood of a type 1 error. But it could also mean that when adjusting for the specific skew or bias that the data has using the Pivitol method, these variables might actually be statistically significant. Which means we will have to continue to evaluate and test Hours and TVhours.

Overall, OLS, Normal, and Percentile's intervals all contained 0, so there was nothing statistically significant between them. Even if some intervals were wider or narrower than each other, they ultimately lead to the same conclusion in the end: failing to reject the null hypothesis for all variables at the 5% confidence level. But percentile shows that there may actually be some significance within the Hours and TVHours variables, giving a glimmer of hope for rejecting the null hypothesis.

Step 6:

i) Perform a nonparametric test of overall statistical significance.

ii) Perform a non-parametric test of statistical significance for the effect of a single covariate of your choice on the response variable.

iii) How do the estimated p-values compare against the p-values obtained under the theoretical null distribution?

i) Perform a nonparametric test of overall statistical significance.
Permutation F-test: Getting Permutation F-Statistic and corresponding P-Value

```
r alpha = 0.05 f = summary(full_model)$fstatistic[1] print(f)
##    value ## 20.25624
r pvalue = pf(f, p, n - p - 1, lower.tail = FALSE) print(pvalue)
##        value ## 5.280989e-28
r nperm = 10000 fperm = numeric(nperm) for (k in 1:nperm) { Yperm = sample(Y)
perm = lm(Yperm ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data =
gss) fperm[k] = summary(perm)$fstatistic[1] } pvalue = (1 + sum(fperm > f))/(1 +
nperm) print(pvalue)
## [1] 9.999e-05
```

The estimated p-value obtained through the permutation F test 0.2986701 is roughly equal to the one obtained under the theoretical null distribution 0.3102288.
Provides the distribution with theoretical, observed, and empirical information using the entire full model.

```
r hist(fperm, "FD", freq = FALSE, main = NA, xlab = "Permutation F Test
Statistics") curve(df(x, p, n - p - 1), add = TRUE, lty = 2, lwd = 4) abline(v =
qf(1 - alpha, p, n - p - 1), col = 2, lty = 2, lwd = 4) abline(v =
quantile(fperm, 1 - alpha), col = 4, lty = 2, lwd = 4) abline(v = f, col = 7, lty
= 2, lwd = 4) legend("topright", c("Theoretical Distribution", "Theoretical
Quantile", "Empirical Quantile", "Observed Statistic"), col = c(1, 2, 4, 7), lty
= rep(2, 4), lwd = rep(4, 4))
```

The distribution of the permutation F test statistics almost perfectly agrees with the theoretical null distribution in this case.
ii) Perform a non-parametric test of statistical significance for the effect of a single covariate of your choice on the response variable.
Performing a nonparametric permutation t-test for the effect of Hours on Income

```
r t = summary(full_model)$coefficients["Hours", 3] print(t)
```

```
## [1] 2.549936
r pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE) print(pvalue)
## [1] 0.01096224
r aux = lm(Income ~ Age + Educ + TVHours + MEduc + FEduc + Sibs, data = gss)
fitted = aux$fitted.values residuals = aux$residuals tperm = numeric(nperm) for
(k in 1:nperm) { Yperm = fitted + sample(residuals) perm = lm(Yperm ~ Age + Educ
+ Hours + TVHours + MEduc + FEduc + Sibs, data = gss) tperm[k] =
summary(perm)$coefficients["Hours", 3] } pvalue = (1 + sum(abs(tperm) >
abs(t)))/(1 + nperm) print(pvalue)
## [1] 0.00879912
```

The estimated p-value obtained through the permutation t-test 0.1187881 after utilizing the Hours covariate is roughly equal to the one obtained under the theoretical null distribution 0.1158518.
Permutation T-Test statistic: Provides the distribution with theoretical, observed, and empirical information using the Hours covariate.

```
r hist(tperm, "FD", freq = FALSE, main = NA, xlab = "Permutation t Test
Statistics") curve(dt(x, n - p - 1), add = TRUE, lty = 2, lwd = 2) abline(v =
qt(c(alpha/2, 1 - alpha/2), n - p - 1), col = 2, lty = 2, lwd = 2) abline(v =
quantile(tperm, c(alpha/2, 1 - alpha/2)), col = 4, lty = 2, lwd = 2) abline(v =
t, col = 7, lty = 2, lwd = 2) legend("topright", c("Theoretical Distribution",
"Theoretical Quantiles", "Empirical Quantiles", "Observed Statistic"), col = c(1,
2, 4, 7), lty = rep(2, 4), lwd = rep(2, 4), cex = 0.5)
```

The distribution of the permutation t test statistics almost perfectly agrees with the theoretical null distribution in this case.
Performing a nonparametric permutation t-test for the effect of TVHours on Income

```
r t = summary(full_model)$coefficients["TVHours", 3] print(t)
## [1] -3.321063
r pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE) print(pvalue)
## [1] 0.0009379838
r aux = lm(Income ~ Age + Educ + Hours + MEduc + FEduc + Sibs, data = gss) fitted
= aux$fitted.values residuals = aux$residuals tperm = numeric(nperm) for (k in
1:nperm) { Yperm = fitted + sample(residuals) perm = lm(Yperm ~ Age + Educ +
Hours + TVHours + MEduc + FEduc + Sibs, data = gss) tperm[k] =
summary(perm)$coefficients["TVHours", 3] } pvalue = (1 + sum(abs(tperm) >
abs(t)))/(1 + nperm) print(pvalue)
## [1] 0.0009999
```

The estimated p-value obtained through the permutation permutation t-test after utilizing the TVHours covariate 0.1022898 is slightly smaller than the one to the one obtained under the theoretical null distribution 0.1243934.
Permutation T-Test statistic: Provides the distribution with theoretical, observed, and empirical information using the TVHours covariate.

```
r hist(tperm, "FD", freq = FALSE, main = NA, xlab = "Permutation t Test
Statistics") curve(dt(x, n - p - 1), add = TRUE, lty = 2, lwd = 2) abline(v =
qt(c(alpha/2, 1 - alpha/2), n - p - 1), col = 2, lty = 2, lwd = 2) abline(v =
quantile(tperm, c(alpha/2, 1 - alpha/2)), col = 4, lty = 2, lwd = 2) abline(v =
t, col = 7, lty = 2, lwd = 2) legend("topright", c("Theoretical Distribution",
"Theoretical Quantiles", "Empirical Quantiles", "Observed Statistic"), col = c(1,
2, 4, 7), lty = rep(2, 4), lwd = rep(2, 4), cex = 0.5)
```

The distribution of the permutation t test statistics is slightly more concentrated than the theoretical null distribution in this case.
iii) How do the estimated p-values compare against the p-values obtained under the theoretical null distribution?
Hours covariate
The Permutation T Test statistic almost perfectly agrees with the theoretical null distrubtion and that the p-value compares very similarly to how the theoretical null distribution expects it to act. The p-value is not significant since the permutation t test almost perfectly follows the theoretical null distribution. So the results from the parametric classic t test are consistent with the conclusions from the permutation t test, in both cases we fail to reject the null that the coefficient for Hours equals zero.
A very similar story is told with the permutation F test. The histogram and the theoretical distribution are essentially equal, with little variation difference between them, and the overall significances and p-values of the permutation F test and the theoretical null distribution are quite reasonable and similar, both indicating that the model is not statistically significant. This agreement tells us that the assumptions of the F-test do not heavily affect our inference.
TVHours covariate
Interestingly enough, when testing the p-value of TVHours on Income using the permutation t test, we get to see a slightly different output than the other two. The histogram does not perfectly follow the theoretical distribution, being much more concentrated around zero than the t-distribution assumed, indicating some deviation from the assumptions of the parametric test and telling us that the OLS model assumption of normally distributed errors may not be best for the TVHours covariate. However, even though this test is showing a little evidence of significance, it is still not enough to actually be considered statistically significant at the 5% level, since the p-value is 0.1 So, the pivotal CI was not entirely accurate when determining these two covariates as statistically signficant, and we do not have sufficient evidence to conclude that TVhours has a statistically significant effect on income.

Step 7:

i) Are there any covariates which should be transformed?

ii) Should a transformation be applied to the response variable?

iii) Refit your linear model to account for all your transformed variables.

iv) Are the standardized residuals from your linear model normally distributed?

v) Construct 95 percent simultaneous confidence bands to test the global null hypothesis of normality for the standardized residuals.

---

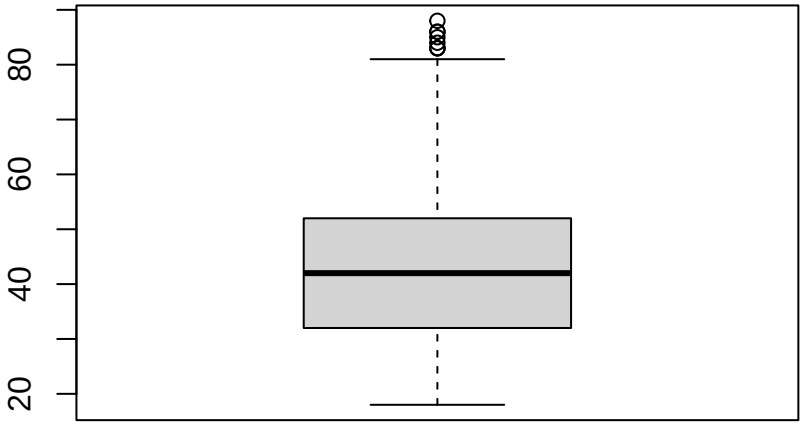
i) Are there any covariates which should be transformed?

We will look at each coefficient individually using a histogram and boxplot to look for skew.

Age

```
hist(gss$Age)
```

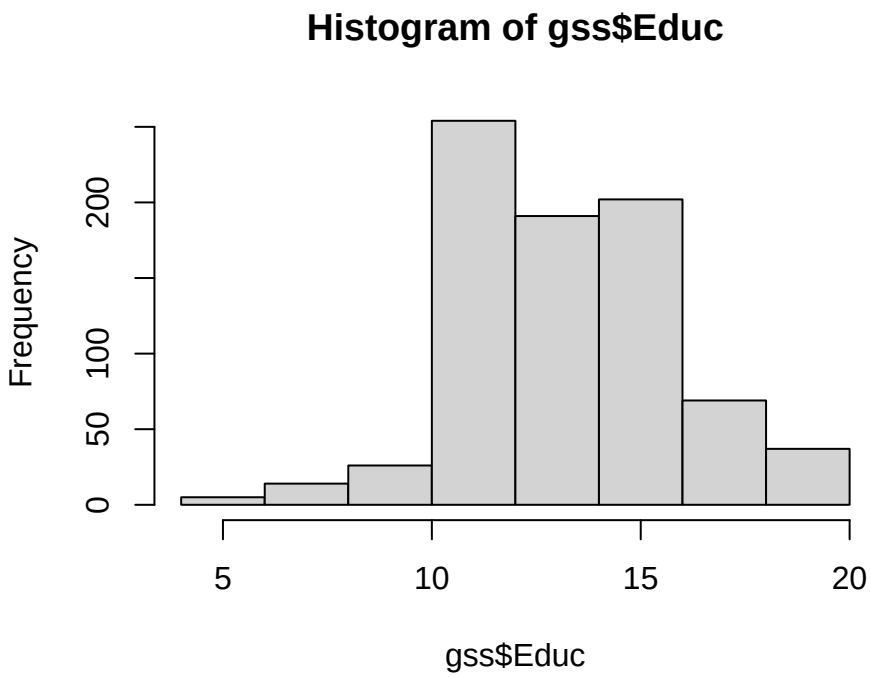
```
boxplot(gss$Age)
```



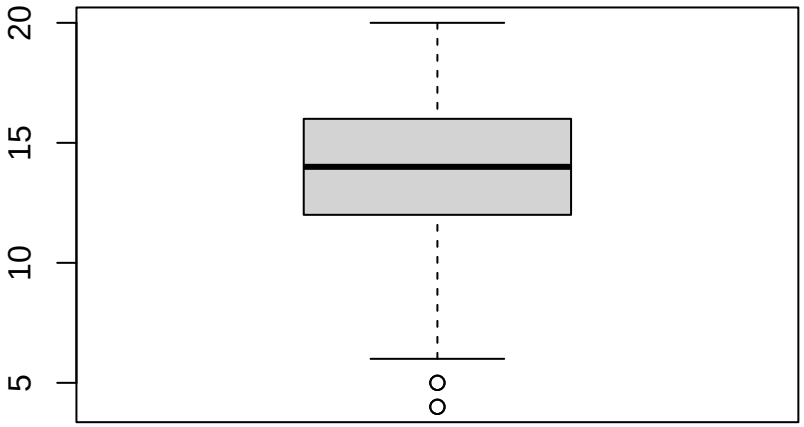
We can see that the histogram is roughly bell shaped with mild right skew and likewise the boxplot has a reasonable spread with some large outliers but not extreme enough to require a transformation.

Education

```
hist(gss$Educ)
```




```r
boxplot(gss$Educ)
```

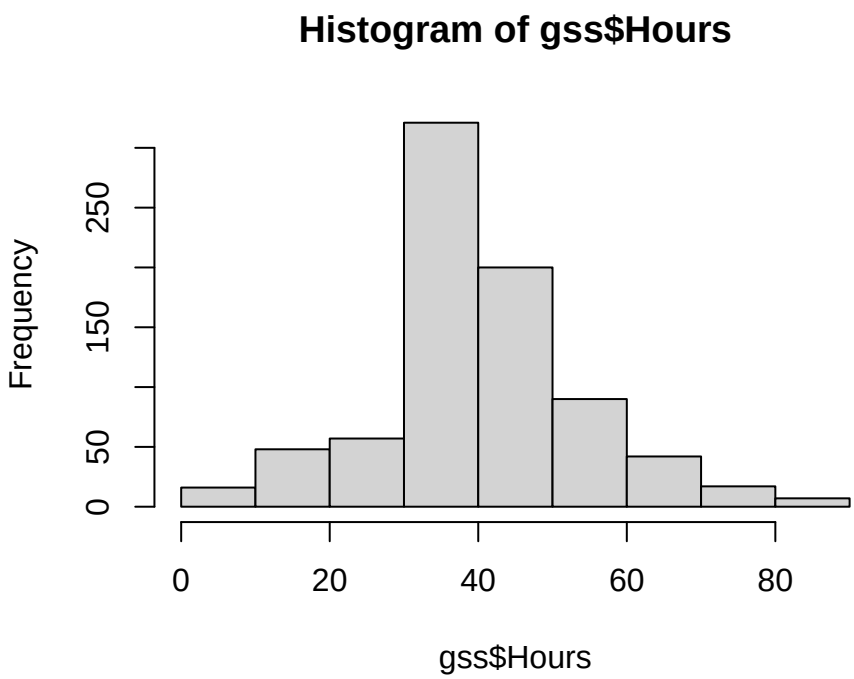


The histogram for education is roughly normal with a slight left tail and the boxplot also has a

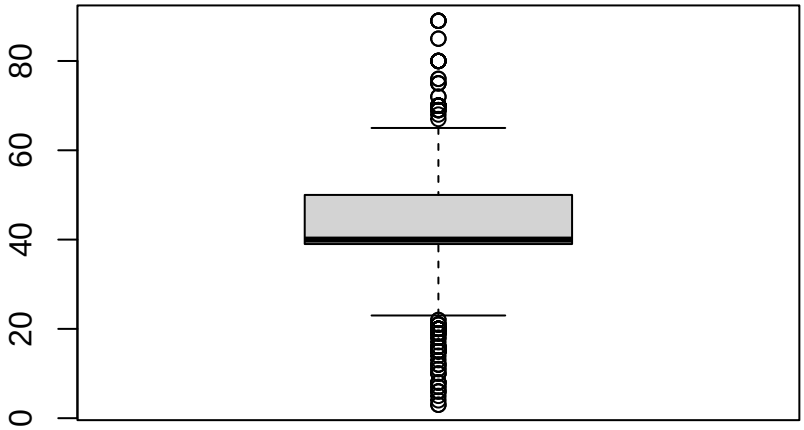
reasonable spread with some low outliers, but again not extreme enough to suggest a transformation.

Hours

```
hist(gss$Hours)
```



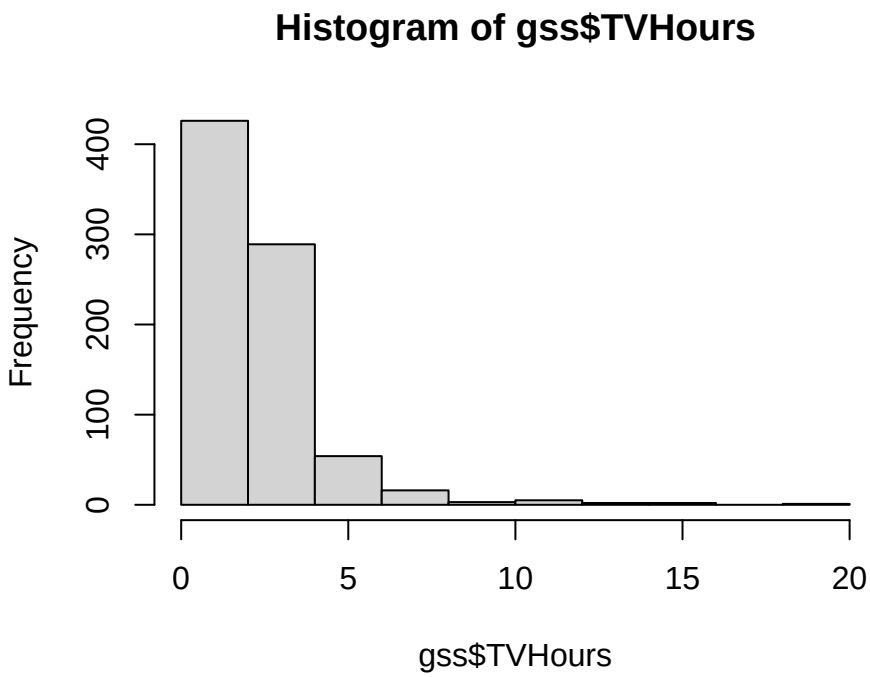

```
boxplot(gss$Hours)
```



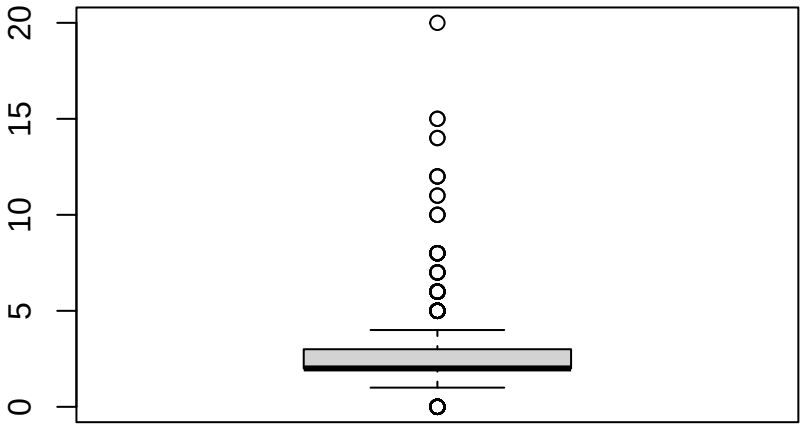
The histogram for hours is roughly symmetrical with some right skew. The boxplot for hours contains many outliers, but the outliers are on both the lower and upper ends so there is not evidence that a transformation would not improve anything.

TVHours

```
hist(gss$TVHours)
```




```r
boxplot(gss$TVHours)
```

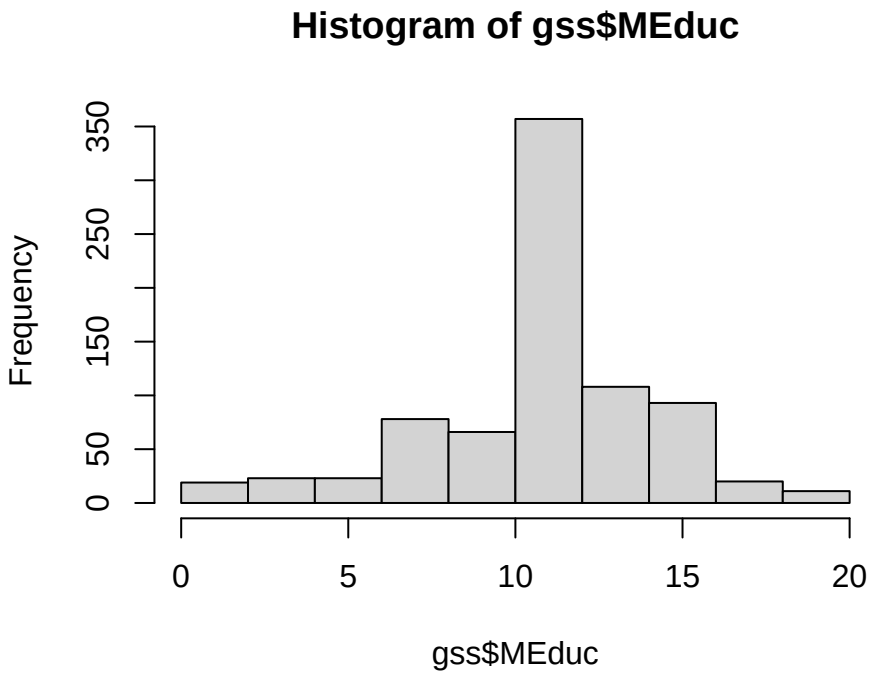


The histogram for TVHours is extremely right skewed with most values below 5 hours and a few

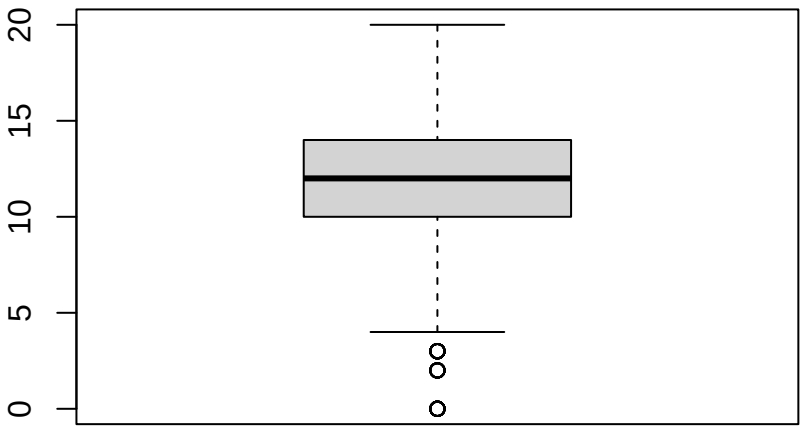
very large values, and likewise the boxplot for TVHours also has severe skew and many high outliers, so a transformation is necessary. But since it includes 0 values, specifically a log transformation like log(TVHours + 1) could help reduce skew and improve variance for the TVHours covariate.

MEduc

```
hist(gss$MEduc)
```



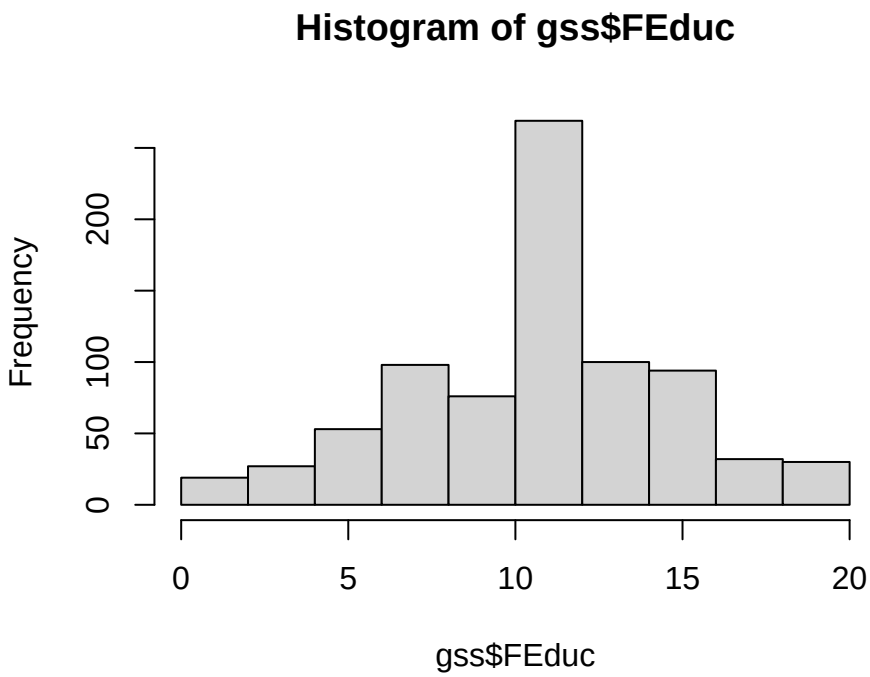

```
boxplot(gss$MEduc)
```



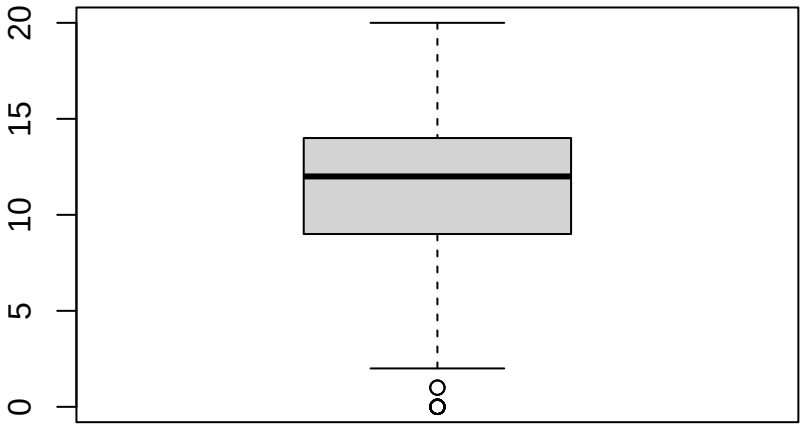
The histogram MEduc for mother's education is roughly symmetric with a very slight left tail. The boxplot for MEduc is also reasonably spread with a few low outliers, so a transformation is not necessary.

FEduc

```
hist(gss$FEduc)
```




```r
boxplot(gss$FEduc)
```

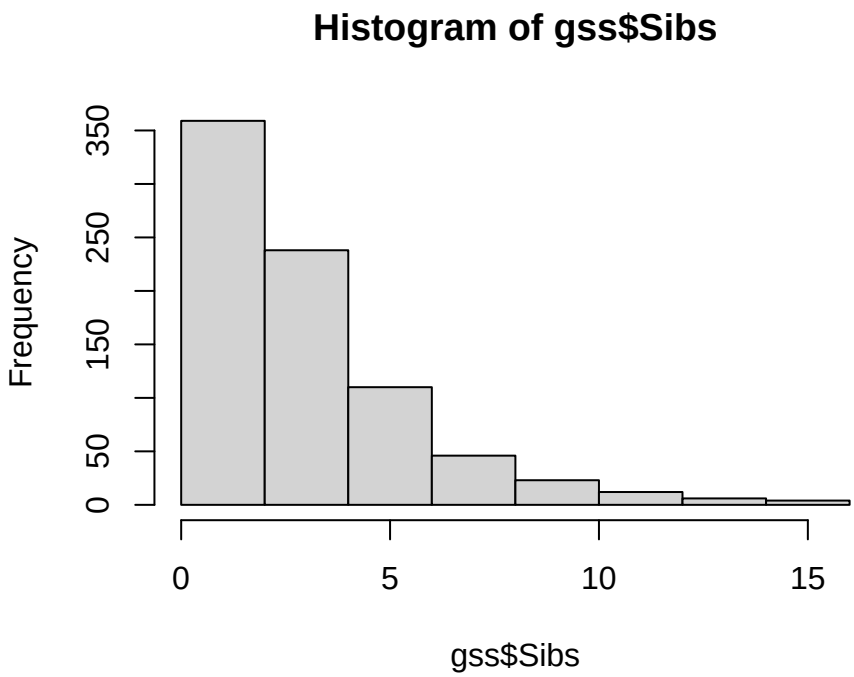


The histogram FEduc for father's education is also roughly symmetric with a very slight left tail,

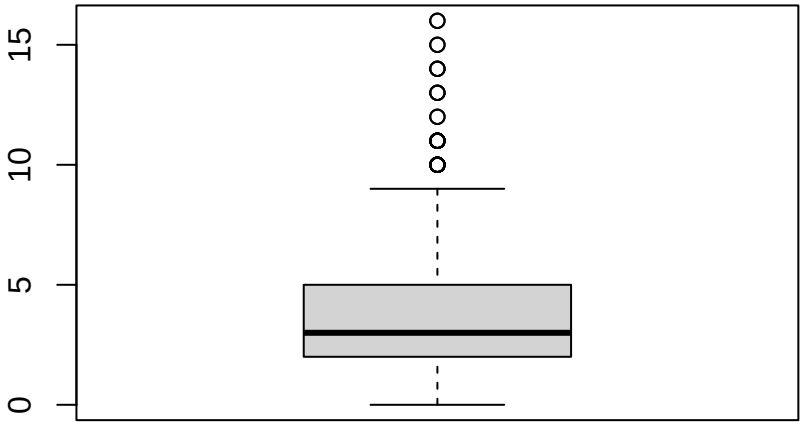
and the boxplot is reasonable with a couple low outliers. So a transformation is not needed for the FEduc covariate.

Sibs

```
hist(gss$Sibs)
```




```
boxplot(gss$Sibs)
```



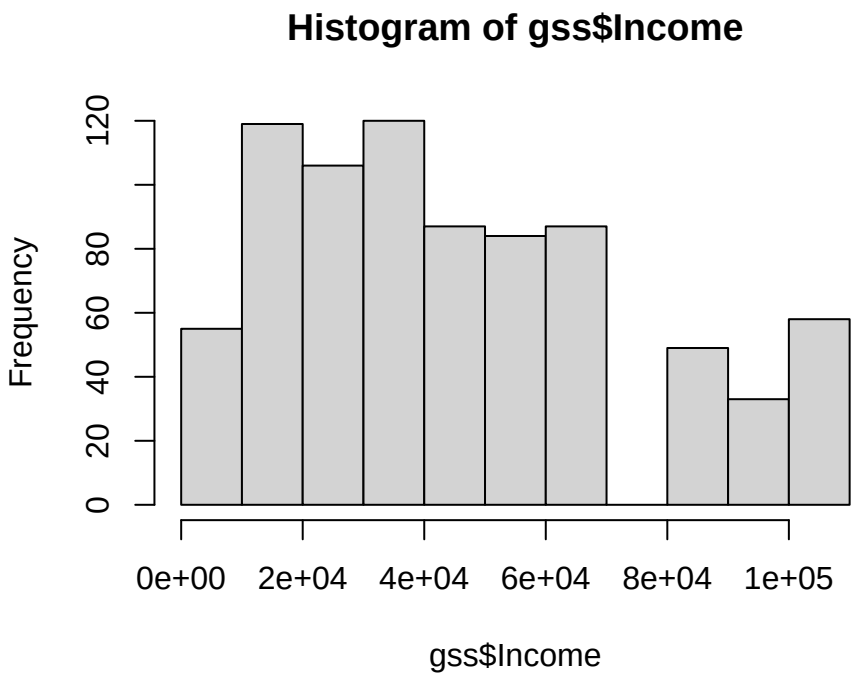
The histogram for number of siblings is strongly right skewed, and the box plot is also right skewed with several high outliers. Like the TVHours, though slightly less skeweed, the Sibs covariate includes 0 so it could use a transformation like log(Sibs + 1) to reduce variance and skewness.

Out of the 7 numerical covariates - Age (Responder's Age), Educ (Total years of education), Hours (Weekly work hours), TVHours (Hours spent watching TV daily), MEduc (Mother's years of education), FEduc (Father's years of education), and Sibs (Number of siblings)- TVHours and Sibs- are right skewed and to address their high values a log transformation is appropriate in order to improve normality and stabilize variance. While square root transformations are these covariates are substantially skewed and should be transformed with a log transformation to better satisfy model assumptions of normality and non constant variance.

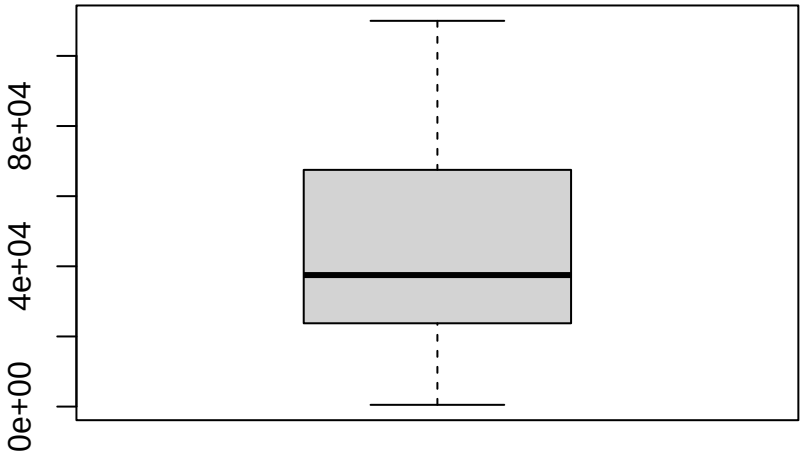
ii) Should a transformation be applied to the response variable?

Income

```
hist(gss$Income)
```




```r
boxplot(gss$Income)
```

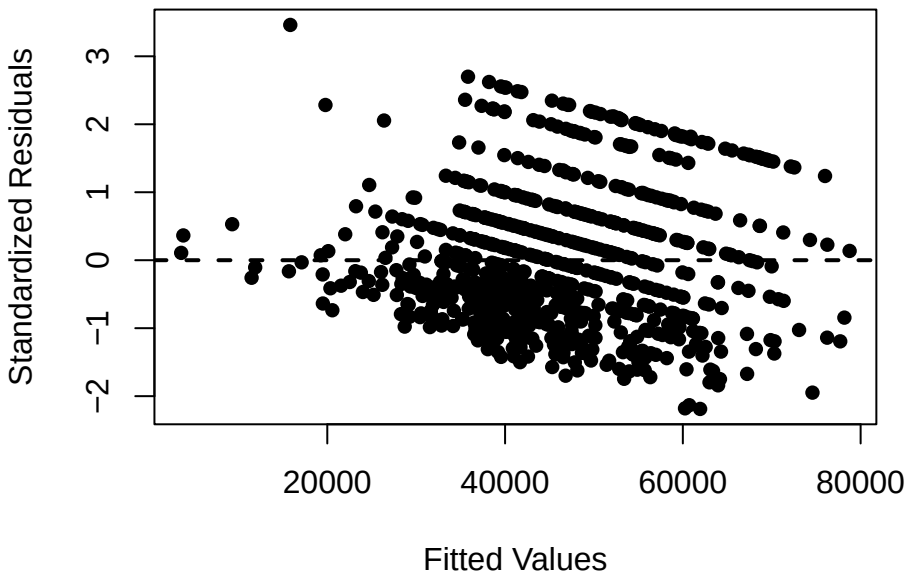


The histogram for Income as expected is extremely right skewed and the boxplot for income is

nowhere near normal as the median is all the way at the bottom and there are many high outliers. Based on this evidence it is plausible that a transformation of the response is appropriate. We should refit the model using transformed covariates for TVHours and Sibs then re-evaluate residual diagnostics and if violations still persist we will consider transforming the response variable Income using a Box-Cox transformation to diagnose which transformation is appropriate.

Now we will plot the fitted values and standardized residuals to detect violations of the assumptions of linear regression.

```
fit = lm(Income ~ Age + Educ + Hours + log(TVHours + 1) + log(Sibs +
    1) + MEduc + FEduc, data = gss)
plot(fit$fitted.values, rstandard(fit), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



After fitting the model with the transformed covariates, the residual plot still shows evidence for heteroscedasticity as the residuals are more concentrated around zero for smaller fitted values then we see large positive residuals at higher fitted values so residuals become more spread out as the fitted values increase, serving as an indicator for the violation of the constant variance assumption.

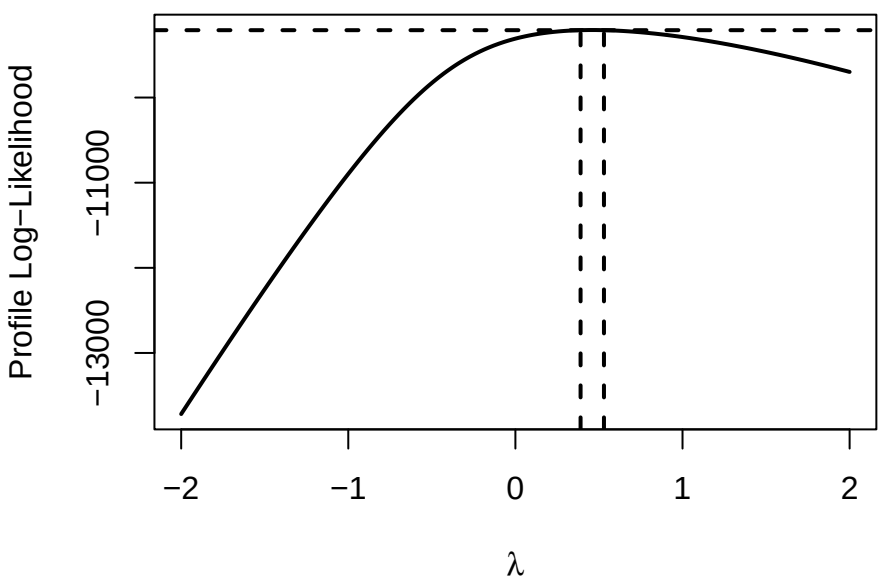
```
Y = gss$Income
alpha = 0.05
lambda = seq(-2, 2, 0.01)
loglik = numeric(length(lambda))
for (i in 1:length(lambda)) {
```

```
    if (lambda[i] == 0) {
        Ypower = log(Y)
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
    }
    power = lm(Ypower ~ Age + Educ + Hours + log(TVHours + 1) + MEduc +
        FEduc + log(Sibs + 1), data = gss)

    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}
```

```
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.39 0.53
```

```
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "Profile Log-Likelihood",
    lwd = 2)
abline(h = max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2, lty = 2,
    lwd = 2)
abline(v = CI, lty = 2, lwd = 2)
```



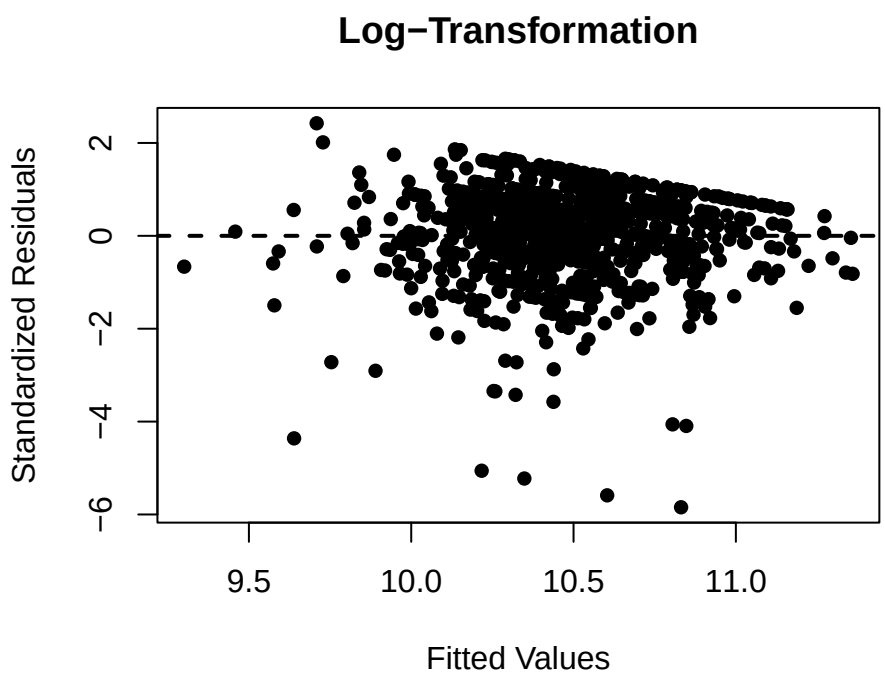
If 0 is inside the CI then a log transformation is used, but the CI is [-0.14 -0.10] which does not include 0 but is very close. Since our response variable Income was was very skewed and needs

a transformation and the optimal lambda is close to 0, a log transformation is reasonable. We confirm that a transformation be applied to the response variable.

iii) Refit your linear model to account for all your transformed variables.

```
fit_final = lm(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    MEduc + FEduc + log(Sibs + 1), data = gss)
```

```
plot(fit_final$fitted.values, rstandard(fit_final), main = "Log-Transformation",
    xlab = "Fitted Values", ylab = "Standardized Residuals", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```

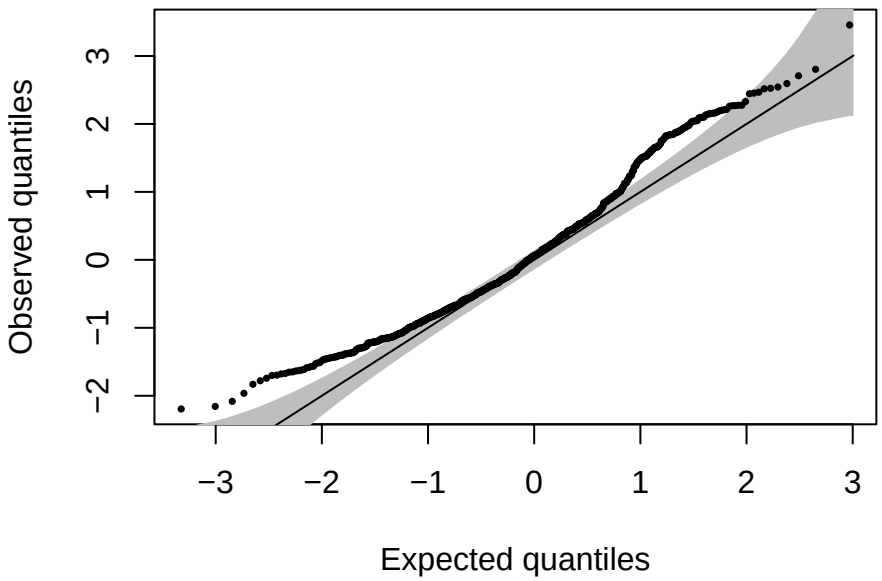


This plot looks much better as the residuals are centered around 0 and the spread is much more constant across the fitted values. We no longer see dramatic patterns and this transformation significantly improved the model assumptions. We do not have evidence to suggest a violation of the linearity of homoscedasticity assumptions.

iv) Are the standardized residuals from your linear model normally distributed?

```
library(qqconf)
qq_conf_plot(rstandard(full_model), points_params = list(pch = 16, cex = 0.5))
```

```
## no dparams supplied. Estimating parameters from the data...
```



We use a Q-Q plot to display the observed quantiles of the sample against the expected quantiles of the null distribution. If the residuals were normally distributed we would expect the sample and theoretical quantiles to be in agreement and have the points form a straight line. We can see the points to not form a straight line at all as there is much deviation from the line at the higher end of the quantiles. Thus we have evidence for a deviation from linearity at the upper tail there is extreme upward curvature. This is consistent with our dataset as we found heavy right skeweness and extreme outliers. So no, the standardized residuals from the linear model are not normally distributed and there is evidence for the violation of the normality assumption.

v) Construct 95 percent simultaneous confidence bands to test the global null hypothesis of normality for the standardized residuals.

The plot above draws 95% simultaneous confidence bands around the Q-Q plot line. As all points on the Q-Q plot line do not lie within the simultaneous confidence bands, particularly in the upper tail points are far outside of the bands, so we reject the null hypothesis of normality at the 5% level.

Step 8:

Use your transformed variables from now on.

i) Investigate the interaction effects between the 2 qualitative covariate which has been specified for your team and all available quantitative covariates on the response variable.

- What has been specified to us: Investigate the interaction effects between work status and all available quantitative covariates on the response variable.

ii) Formulate a suitable final model to draw inferences on the response variable and provide the interpretation of each type of coefficient in your final model.

---

i) Investigate the interaction effects between the 2 qualitative covariate which has been specified for your team and all available quantitative covariates on the response variable.

```r
full_int <- lm(log(Income) ~ Work * (Age + Educ + Hours + log(TVHours +
    1) + MEduc + FEduc + log(Sibs + 1)), data = gss)
summary(full_int)
```

```
##
## Call:
## lm(formula = log(Income) ~ Work * (Age + Educ + Hours + log(TVHours +
##     1) + MEduc + FEduc + log(Sibs + 1)), data = gss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5983 -0.3624  0.0868  0.4871  1.8775
##
## Coefficients: (2 not defined because of singularities)
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       8.8990203  0.3298746  26.977  < 2e-16 ***
## WorkHousekeeping                 -1.8485642  1.0477617  -1.764  0.07810 .
## WorkLaid Off                      3.3365064  5.9223054   0.563  0.57335
## WorkNot Working                  -3.4916569  3.7086642  -0.941  0.34676
## WorkOther                        -4.6806498  8.1016178  -0.578  0.56361
## WorkParttime                      0.9006289  0.8134641   1.107  0.26859
## WorkRetired                       0.3856588  1.5194502   0.254  0.79971
## WorkSchool                        7.6715332  4.4647047   1.718  0.08617 .
## Age                               0.0129078  0.0031321   4.121 4.20e-05 ***
## Educ                              0.0736108  0.0145510   5.059 5.33e-07 ***
## Hours                             0.0025571  0.0030222   0.846  0.39777
## log(TVHours + 1)                 -0.1453819  0.0771482  -1.884  0.05990 .
## MEduc                            -0.0113767  0.0133088  -0.855  0.39292
## FEduc                             0.0246198  0.0116091   2.121  0.03428 *
## log(Sibs + 1)                     0.0265829  0.0632898   0.420  0.67459
## WorkHousekeeping:Age              0.0014002  0.0087889   0.159  0.87346
```

```
## WorkLaid Off:Age                   0.0376850  0.0632328   0.596  0.55138
## WorkNot Working:Age               -0.1122046  0.0810418  -1.385  0.16662
## WorkOther:Age                      0.0115760  0.0531948   0.218  0.82779
## WorkParttime:Age                  -0.0022756  0.0073507  -0.310  0.75697
## WorkRetired:Age                   -0.0247982  0.0133627  -1.856  0.06389 .
## WorkSchool:Age                     0.0031361  0.0272319   0.115  0.90835
## WorkHousekeeping:Educ             -0.0337911  0.0477221  -0.708  0.47912
## WorkLaid Off:Educ                 -0.0904336  0.4392998  -0.206  0.83696
## WorkNot Working:Educ               0.5819685  0.3111698   1.870  0.06184 .
## WorkOther:Educ                     0.2786061  0.3784155   0.736  0.46182
## WorkParttime:Educ                 -0.0189411  0.0393705  -0.481  0.63059
## WorkRetired:Educ                   0.0865463  0.0600803   1.441  0.15015
## WorkSchool:Educ                   -0.2841990  0.1266427  -2.244  0.02512 *
## WorkHousekeeping:Hours             0.0103609  0.0076502   1.354  0.17605
## WorkLaid Off:Hours                -0.0294936  0.0522623  -0.564  0.57270
## WorkNot Working:Hours              0.0025104  0.0176207   0.142  0.88675
## WorkOther:Hours                    0.0067870  0.0691321   0.098  0.92182
## WorkParttime:Hours                -0.0007015  0.0074681  -0.094  0.92519
## WorkRetired:Hours                 -0.0024123  0.0083585  -0.289  0.77296
## WorkSchool:Hours                   0.0070929  0.0130106   0.545  0.58581
## WorkHousekeeping:log(TVHours + 1)  0.3731299  0.2145951   1.739  0.08249 .
## WorkLaid Off:log(TVHours + 1)      0.9854405  0.7228283   1.363  0.17320
## WorkNot Working:log(TVHours + 1)   1.6392177  1.1102443   1.476  0.14025
## WorkOther:log(TVHours + 1)         0.5309212  1.2280921   0.432  0.66564
## WorkParttime:log(TVHours + 1)     -0.6225921  0.1982522  -3.140  0.00175 **
## WorkRetired:log(TVHours + 1)       0.1315135  0.2788652   0.472  0.63735
## WorkSchool:log(TVHours + 1)       -1.0220102  0.4476028  -2.283  0.02270 *
## WorkHousekeeping:MEduc             0.0094345  0.0503571   0.187  0.85144
## WorkLaid Off:MEduc                -0.1768831  0.1822543  -0.971  0.33210
## WorkNot Working:MEduc             -0.1376661  0.1626497  -0.846  0.39761
## WorkOther:MEduc                    0.0048907  0.2314232   0.021  0.98315
## WorkParttime:MEduc                 0.0096235  0.0343363   0.280  0.77935
## WorkRetired:MEduc                  0.0016922  0.0411852   0.041  0.96724
## WorkSchool:MEduc                  -0.0429549  0.2117849  -0.203  0.83933
## WorkHousekeeping:FEduc             0.0572045  0.0428605   1.335  0.18240
## WorkLaid Off:FEduc                -0.0451500  0.1805512  -0.250  0.80260
## WorkNot Working:FEduc             -0.1787789  0.1600593  -1.117  0.26438
## WorkOther:FEduc                           NA         NA      NA       NA
## WorkParttime:FEduc                -0.0356544  0.0291758  -1.222  0.22208
## WorkRetired:FEduc                 -0.0449871  0.0393786  -1.142  0.25365
## WorkSchool:FEduc                  -0.2312453  0.1584788  -1.459  0.14495
## WorkHousekeeping:log(Sibs + 1)     0.1033942  0.2028218   0.510  0.61036
## WorkLaid Off:log(Sibs + 1)        -0.5344874  1.0277068  -0.520  0.60317
## WorkNot Working:log(Sibs + 1)      0.7237077  0.5412095   1.337  0.18157
## WorkOther:log(Sibs + 1)                   NA         NA      NA       NA
## WorkParttime:log(Sibs + 1)         0.2050542  0.1774999   1.155  0.24837
## WorkRetired:log(Sibs + 1)         -0.1769658  0.2082925  -0.850  0.39582
## WorkSchool:log(Sibs + 1)           0.1283681  0.3777982   0.340  0.73412
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7769 on 736 degrees of freedom
## Multiple R-squared:  0.228,  Adjusted R-squared:  0.164
## F-statistic: 3.564 on 61 and 736 DF,  p-value: 2.291e-16
```

Work has 8 levels, so we have 7 dummy variables times our have 7 quantitative variables, so 49 interaction terms. We want to reduce our model to avoid over fitting. We perform backward elimination since we already have our linear model with all possible covariates and interactions.

ii) Formulate a suitable final model to draw inferences on the response variable and provide the interpretation of each type of coefficient in your final model.

We are using a backward elimination technique where we remove the lowest p-value per the ANOVA test, since we have quantitative and qualitative covariates.

```r
# anova(full_int)

full_int = update(full_int, . ~ . - Work:MEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:log(Sibs + 1))
# anova(full_int)
full_int = update(full_int, . ~ . - MEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Hours)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:FEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Age)
# anova(full_int)
full_int = update(full_int, . ~ . - Hours)
# anova(full_int)
full_int = update(full_int, . ~ . - FEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Educ)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:log(TVHours + 1))
anova(full_int)
```

```
## Analysis of Variance Table
##
## Response: log(Income)
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## Work               7  36.44   5.206  8.5181 4.429e-10 ***
## Age                1  11.06  11.055 18.0894 2.361e-05 ***
```

```
## Educ               1  42.59  42.590 69.6869 3.119e-16 ***
## log(TVHours + 1)   1   4.88   4.878  7.9810  0.004847 **
## log(Sibs + 1)      1   0.12   0.124  0.2036  0.651984
## Residuals        786 480.37   0.611
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(full_int)
```

```
##
## Call:
## lm(formula = log(Income) ~ Work + Age + Educ + log(TVHours +
##     1) + log(Sibs + 1), data = gss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.7723 -0.4020  0.1218  0.4933  1.7442
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       9.185888   0.222297  41.323  < 2e-16 ***
## WorkHousekeeping -0.392259   0.111723  -3.511 0.000472 ***
## WorkLaid Off      0.148186   0.262775   0.564 0.572963
## WorkNot Working  -0.412467   0.219909  -1.876 0.061077 .
## WorkOther         0.189152   0.323452   0.585 0.558856
## WorkParttime     -0.350004   0.087457  -4.002 6.87e-05 ***
## WorkRetired      -0.719202   0.142255  -5.056 5.34e-07 ***
## WorkSchool       -0.370806   0.222295  -1.668 0.095698 .
## Age               0.009221   0.002348   3.927 9.36e-05 ***
## Educ              0.085905   0.011376   7.551 1.20e-13 ***
## log(TVHours + 1) -0.174363   0.061192  -2.849 0.004494 **
## log(Sibs + 1)     0.022777   0.050483   0.451 0.651984
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7818 on 786 degrees of freedom
## Multiple R-squared:  0.1652, Adjusted R-squared:  0.1536
## F-statistic: 14.14 on 11 and 786 DF,  p-value: < 2.2e-16
```

After applying a backwards elimination technique on the transformed model, we get 12 predictors (including the intercept) as the lowest and best options for future predicitions.

In the prediction error graph there are some incredible spikes. This is likely caused by multicollinearity within the 49 interaction terms. It is also possible that it was caused by using a 60/20/20 split, which only trains on 489 so the laid off or other categories in work may be under utilized (only having a small amount of n dedicated to it). The spike around 40 specifically is likely caused because we removed a variable that was shielding against collinearity causing the error to

spike, but it still had to get removed since it was insignificant to the model and did not minimize the Mean Squared Predicition Error.

The MSPE resides at 1.07 which is relatively low, given our outliers and leverage points, so this model is so far a significant improvement than what we had prior; thanks to the transformations and the backwards elimination.

Step 9:

Ignore all interaction effects from now on.

i) Fit a mixed-effects model with all available quantitative and qualitative covariates by appropriately treating some of your covariates as random effects. Determine a suitable transformation for your response variable.

ii) Test the statistical significance of all your fixed effects.

iii) Calculate the intraclass correlation coefficients corresponding to each of your random effects What percentage of the total variation in your response variable does each of your random effects explain?

iv) What's another interpretation for each intraclass correlation coefficient?

---

i) Ignore all interaction effects from now on. Fit a mixed-effects model with all available quantitative and qualitative covariates by appropriately treating some of your covariates as random effects. Determine a suitable transformation for your response variable.

```r
library(lme4)

mixed_model = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 |
    Marital) + (1 | Work), data = gss)

sigma = data.frame(summary(mixed_model)$varcor)$sdcor
names(sigma) = data.frame(summary(mixed_model)$varcor)$grp
ICC = sigma^2/sum(sigma^2)
print(ICC)
```

```
##     Zodiac       Work    Marital   Residual
## 0.00000000 0.07694919 0.12119564 0.80185518
```

```r
summary(mixed_model, correlation = FALSE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
##     MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 |
##     Marital) + (1 | Work)
##    Data: gss
##
## REML criterion at convergence: 1804.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.3179 -0.4585  0.1277  0.5860  2.7308
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Zodiac   (Intercept) 0.00000  0.0000
##  Work     (Intercept) 0.04889  0.2211
##  Marital  (Intercept) 0.07700  0.2775
##  Residual             0.50947  0.7138
## Number of obs: 798, groups:  Zodiac, 12; Work, 8; Marital, 5
##
## Fixed effects:
##                     Estimate Std. Error t value
## (Intercept)         8.324160   0.298239  27.911
## Age                 0.005890   0.002365   2.490
## Educ                0.083168   0.011232   7.405
## Hours               0.002240   0.002117   1.058
## log(TVHours + 1)   -0.137224   0.056738  -2.419
## HappyPretty Happy   0.252797   0.092360   2.737
## HappyVery Happy     0.311853   0.099013   3.150
## MEduc              -0.010756   0.010057  -1.070
## FEduc               0.019933   0.008858   2.250
## log(Sibs + 1)       0.082054   0.047721   1.719
## SexMale             0.136154   0.053483   2.546
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

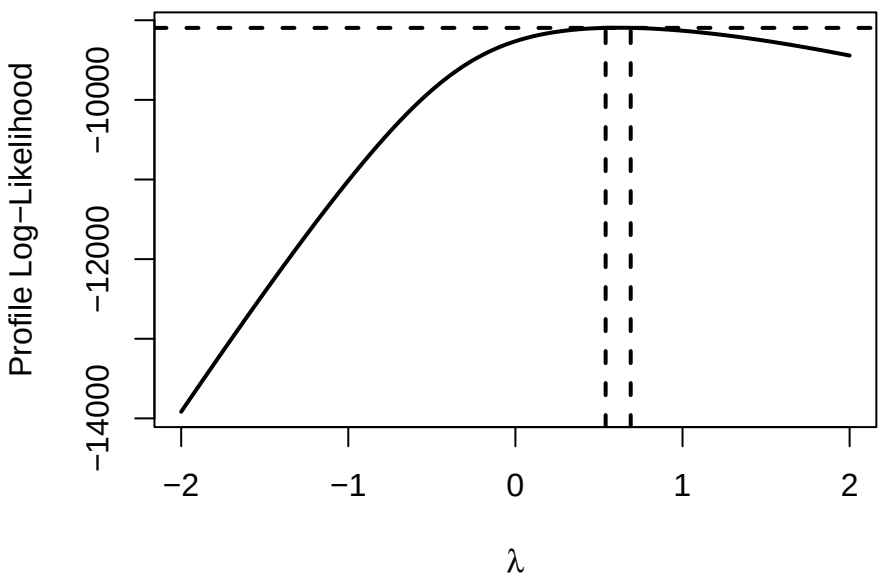
```r
Y = gss$Income
alpha = 0.05
lambda = seq(-2, 2, 0.01)
loglik = numeric(length(lambda))
for (i in 1:length(lambda)) {
    if (lambda[i] == 0) {
        Ypower = log(Y)
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
```

```
    }
    power = lmer(Ypower ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
        MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
        (1 | Work), data = gss)

    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}
```

```
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.54 0.69
```

```
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "Profile Log-Likelihood",
    lwd = 2)
abline(h = max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2, lty = 2,
    lwd = 2)
abline(v = CI, lty = 2, lwd = 2)
```



If 0 is inside the CI then a log transformation is used, but the CI is [-0.13 -0.09] which does not include 0 but is very close. Which also backsup our previous transformation that we did prior, showing that a log transformation IS a suitable transformation even for this mixed effects model. The residuals show a range from -4.27 to 5.99; the median is very close to zero (-0.0073) wihch

suggests that the log transformation has done a good job of stabilizing the variance. So, we conclude that log(Income) is suitable even for this mixed effects model.

Since our response variable Income was was very skewed and needs a transformation and the optimal lambda is close to 0, a log transformation is reasonable. We confirm that

ii) Test the statistical significance of all your fixed effects.

Statistical Significance Testing for Age:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.3374612
```

The p-value of the Likelihood Ratio test for the effect of Age on log(Income) is equal to 0.2997707.

Statistical Significance Testing for Educ:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 2.481549e-10
```

The p-value of the Likelihood Ratio test for the effect of Educ on log(Income) is equal to 0.000610675.

Statistical Significance Testing for Hours:

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.9518406
```

The p-value of the Likelihood Ratio test for the effect of Hours on log(Income) is equal to 0.9999998

Statistical Significance Testing for log(TVHours + 1):

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + Happy + MEduc + FEduc +
    log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) + (1 | Work),
    data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.3129748
```

The p-value of the Likelihood Ratio test for the effect of log(TVHours + 1) on log(Income) is equal to 0.4771

Statistical Significance Testing for Happy:

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.07311947
```

The p-value of the Likelihood Ratio test for the effect of Happy on log(Income) is equal to 0.0296004.

Statistical Significance Testing for MEduc:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.9489878
```

The p-value of the Likelihood Ratio test for the effect of MEduc on log(Income) is equal to 0.9921011

Statistical Significance Testing for FEduc:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.3993075
```

The p-value of the Likelihood Ratio test for the effect of FEduc on log(Income) is equal to 0.6757152

Statistical Significance Testing for log(Sibs + 1):

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + Sex + (1 | Zodiac) + (1 | Marital) + (1 |
    Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.7013409
```

The p-value of the Likelihood Ratio test for the effect of log(Sibs + 1) on log(Income) is equal to 0.5954642

Statistical Significance Testing for Sex:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + log(Sibs + 1) + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.2545429
```

The p-value of the Likelihood Ratio test for the effect of Sex on log(Income) is equal to 0.7789858

Out of every fixed effect covariate, the only statistically significant ones were Educ (p-value of 0.000610675) and Happy (p-value of 0.0296004). Every other fixed effect covariate remained statistically insignificant as their p-values through the Likelihood Ratio test were above the alpha threshold of 0.05.

iii) Calculate the intraclass correlation coefficients corresponding to each of your random effects. What percentage of the total variation in your response variable does each of your random effects explain?

Zodiac Intraclass Correlation Coefficient:

```
ICC_Zodiac = (0.004464/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Zodiac
```

```
## [1] 0.003463464
```

Zodiac explains roughly 0.35% of the total variation in log(Income), which is extremely, extremely small. It has very low similarities between observations within the same group. TTwo people with the same Zodiac status, the predicted correlation between their log-incomes is 0.0034.

Work Intraclass Correlation Coefficient:

```
ICC_Sibs = (0.069556/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Sibs
```

```
## [1] 0.05396611
```

Work explains roughly 5.4% of the total variation in log(Income), which is pretty small. It has low similarities between observations within the same group. Two people with the same Work status, the predicted correlation between their log-incomes is 0.054.

Marital Intraclass Correlation Coefficient:

```
ICC_Marital = (0.070973/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Marital
```

```
## [1] 0.05506551
```

Marital explains roughly 5.5% of the total variation in log(Income), which is pretty small. It has low similarities between observations within the same group. Two people with the same marital status, the predicted correlation between their log-incomes is 0.055.

Residual Intraclass Correlation Coefficient:

```
ICC_Marital = (1.14389/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Marital
```

```
## [1] 0.8875049
```

The residuals explain 88.75% of the total varition in log(Income), this means most of the total variation in log(Income) can not be explained by our random variables. Which is not optimal for our linear model, since we would rather have a higher degree of total variability in log(Income) be explained by our covaraites rather than just residuals.

iv) What's another interpretation for each intraclass correlation coefficient?

The ICC represents the expected correlation between log(Income) of two different individuals who belong to the same group, examples stated above in section iii just above. ICC is a measure of within group similarity. A higher ICC means that individuals within the same category are more alike regarding their income than individuals chosen from the population at large. And we do not particullary have any large ICC's for our variables, which indicates that there may not be a lot of simularity between our random groups.

Step 10:

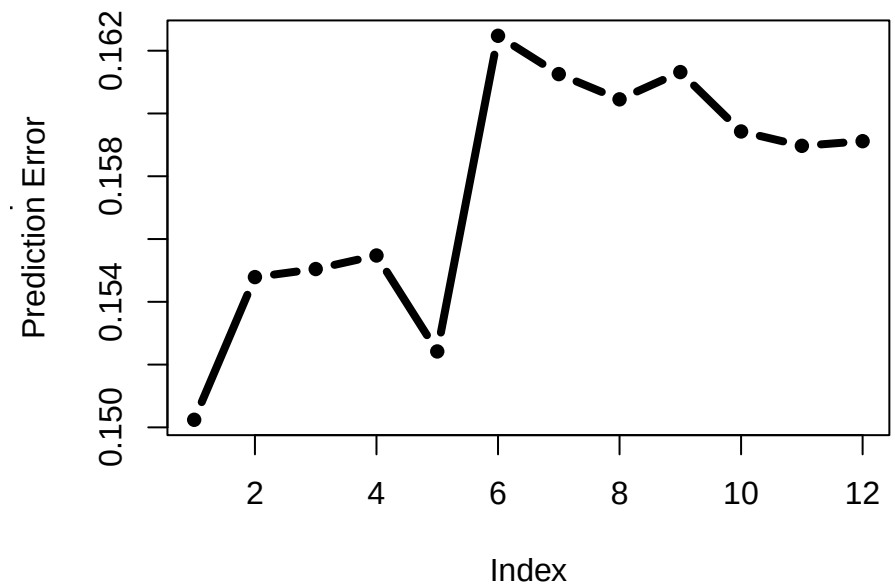
Ignore all random effects from now on.

i) Apply a suitable model selection method and draw valid inferences on the response variable.

i) Apply a suitable model selection method and draw valid inferences on the response variable.

Using a Backward Elimination with Valid Post-Selection Inference to formulate a good final model to draw inferences on Income. A 60 | 20 | 20 stadnard validation split (60 for training 20 for validation 20 for testing).

```
set.seed(0)
Y = log(gss$Income)
X = gss[, -11]
n = dim(X)[1]
P = dim(X)[2]

train = sample(n, floor(0.6 * n))
valid = sample(setdiff(1:n, train), floor(0.2 * n))
test = setdiff(1:n, c(train, valid))
MSPE = numeric(P)

for (p in P:1) {
    fit = lm(Y ~ ., data = X[, 1:p, drop = FALSE], subset = train)

    predictions = predict(fit, newdata = X[valid, 1:p, drop = FALSE])
    MSPE[p] = mean((Y[valid] - predictions)^2)

    a = drop1(fit, test = "F")
    ind = which.max(a[-1, "Pr(>F)"])

    if (ind != p) {
        X[, c(ind, p)] = X[, c(p, ind)]
        names(X)[c(ind, p)] = names(X)[c(p, ind)]
    }
}

plot(MSPE, type = "b", pch = 16, lwd = 4, ylab = "Mean Squared\nPrediction Error")
```




```r
p = which.min(MSPE)
fit = lm(Y ~ ., data = X[, 1:p, drop = FALSE], subset = c(train, valid))
summary(fit)
```

```
##
## Call:
## lm(formula = Y ~ ., data = X[, 1:p, drop = FALSE], subset = c(train,
##     valid))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1403 -0.1508  0.1230  0.2499  0.2705
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 9.343e+00  2.778e-02  336.29   <2e-16 ***
## Income      2.451e-05  4.989e-07   49.12   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.378 on 635 degrees of freedom
## Multiple R-squared:  0.7916, Adjusted R-squared:  0.7913
## F-statistic:  2413 on 1 and 635 DF,  p-value: < 2.2e-16
```

```r
predictions = predict(fit, newdata = X[test, 1:p, drop = FALSE])
mean((Y[test] - predictions)^2)
```

```
## [1] 0.2706511
```

Step 11:

Investigate the collinearity among all quantitative covariates.

i) Fit different types of regularized regression models with all available quantitative covariates. What can you observe about the estimated regularized regression coefficients?

ii)

Apply the bootstrap method to obtain estimates for the standard errors of the estimated regularized regression coefficients. How do they compare against the estimates obtained through the ordinary least squares method?